

(1, 1, 6),(3, 3, 2) 9 cases

(1, 1, 5),(3, 3, 1) 9 cases

(1, 1, 4), 3 cases

(1, 1, 3)

(2, 2, 1), 3 cases

(1, 1, 2), 3 cases

Total cases of isosceles points which are not equilateral = $15 + 12 + 9 + 9 + 9 + 3 + 6 + 3 = 66$

INTEGER TYPE QUESTIONS

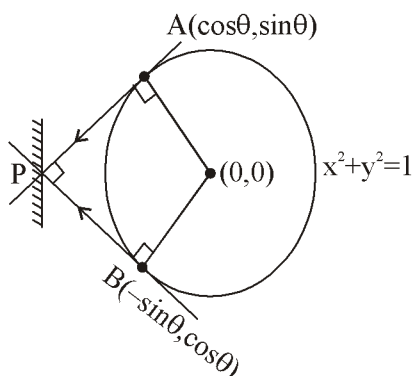
C01/4/153. A ray emitted from the point $A(\cos\theta, \sin\theta)$ strikes a mirror at a variable point P , then reflects and passes through a point $B(-\sin\theta, \cos\theta)$ such that incident and reflected rays are tangents to the circle $x^2 + y^2 = 1$, then locus of the variable point is $x^2 + y^2 = k$, then k is equal to _____.

बिन्दु $A(\cos\theta, \sin\theta)$ से उत्सर्जित एक किरण दर्पण पर स्थित चर बिन्दु P से टकराकर परावर्तित होती है, और बिन्दु $B(-\sin\theta, \cos\theta)$ से होकर गुजरती है। आपतित और परावर्तित किरणें वृत्त $x^2 + y^2 = 1$ को स्पर्श करती हैं। तो चर बिन्दु का बिन्दुपथ $x^2 + y^2 = k$ होगा, तो k का मान है _____.

Ans.

2

Sol.



It is obvious that tangents are perpendicular, then

locus of P is its director

circle $x^2 + y^2 = 2$.

C01/4/154. A ray of light incident at the point $(-3, -2)$ gets reflected from the tangent at $(0, -2)$ to the circle $x^2 + y^2 = 4$. The reflected ray touches the circle. The equation of the line along which the incident ray moved is $Jx - Ky + 46 = 0$. The value of $J + K$ is _____.

एक प्रकाश किरण बिन्दु $(-3, -2)$ पर आपतित होती है और वृत्त $x^2 + y^2 = 4$ पर स्थित बिन्दु $(0, -2)$ पर स्पर्शरेखा से परावर्तित होती है। परावर्तित किरण इस वृत्त को स्पर्श करती है। यदि आपतित किरण का समीकरण $Jx - Ky + 46 = 0$ है, तो $J + K$ का मान क्या होगा _____.

Ans. 7

Sol. Any line through $P(-3, -2)$ is $y + 2 = m(x + 3)$

It touches the circle if $mx - y + 3m - 2 = 0$

$$\left| \frac{3m - 2}{\sqrt{1 + m^2}} \right| = 2$$

$$9m^2 + 4 - 12 = 4 + 4m^2$$

$$5m^2 - 12m = 0$$

$$m = 0, \frac{12}{5}$$

Equation of PB is

$$y + 2 = \frac{12}{5}(x + 3)$$

$$5y + 10 = 12x + 36$$

$$12x - 5y + 26 = 0$$

$$\text{Slope } m = \frac{12}{5}$$

$$\tan \theta = \left| \frac{\frac{12}{5} - 0}{1 + 0} \right|$$

$$\tan \theta = \frac{12}{5}$$

$$\angle CPB = \angle DPA$$

$$\therefore \angle CPA = \theta - \theta$$

$$\tan(\pi - \theta) = -\frac{12}{5}$$

Let slope of AP is m_1

$$m_1 = -\frac{12}{5}$$

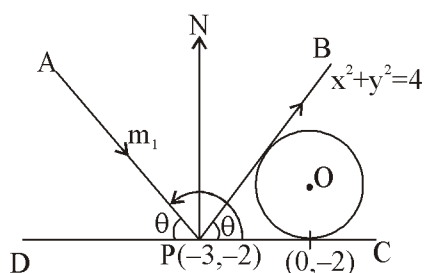
So equation of AP is

$$y + 2 = -\frac{12}{5}(x + 3)$$

$$5y + 10 = -12x - 36$$

$$12x + 5y + 46 = 0$$

$$Jx - ky + 46 = 0$$



$$J = 12 \quad K = -5$$

$$J + K = 7$$

C01/4/155. The line $L_1 : 3x + 4y + 12 = 0$ is rotated through an angle of $\tan^{-1}\left(\frac{24}{7}\right)$ in anticlockwise about the point where it intersects the x-axis, line obtained in new position is denoted as $L_2 = 0$. If incentre of the triangle formed by $L_1 = 0$, $L_2 = 0$ and y-axis is $(4\lambda, 3k)$, then the value of $10k - 24\lambda$ is _____.

रेखा $L_1 : 3x + 4y + 12 = 0$ को वामावर्त दिशा में $\tan^{-1}\left(\frac{24}{7}\right)$ कोण से उस बिन्दु के सापेक्ष घुमाया जाता है जहाँ रेखा L_1 , x-अक्ष को काटती है तो रेखा $L_2 = 0$ प्राप्त होती है। यदि $L_1 = 0$, $L_2 = 0$ और y-अक्ष से बनी त्रिभुज का अंतःकेन्द्र $(4\lambda, 3k)$ है, तो $10k - 24\lambda$ का मान है _____.

Ans. 9

Sol. $\theta = \tan^{-1}\left(\frac{24}{7}\right)$

$$\frac{24}{7} = \left(\frac{m - (-3/4)}{1 + m(-3/4)} \right)$$

$$m = \frac{3}{4}$$

$\therefore$ equation of the line $L_2 = 0$ is

$$L_2 : 3x - 4y + 12 = 0$$

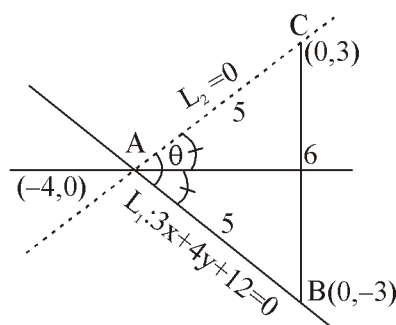
Now, incentre of the ΔABC is

$$\left(\frac{5 \times 0 + 6 \times (-4) + 5 \times 0}{16}, \frac{5 \times 3 + 6 \times 0 + 5 \times (-3)}{16} \right)$$

$$\equiv \left(-\frac{3}{2}, 0 \right) \equiv (4\lambda, 3k)$$

$$\therefore \lambda = -\frac{3}{8}, k = 0$$

$$10k - 24\lambda = 9$$



C01/4/156. L_1, L_2, L_3 are three lines where $L_1 \equiv 2x + y = 2$, $L_2 \equiv x - 2y = -1$ and L_3 is any line passing through the point of intersection of L_1 and L_2 . A line L passing through $P(0, 1)$ intersects L_1, L_2 and L_3 at A, B and C respectively. If intercept of L between L_1 and L_2 subtends right angle at origin and

$$\left(\frac{1}{PA} \right)^2 + \left(\frac{1}{PB} \right)^2 + PC^2 = 5, \text{ then find } L_3 \text{ and } L \text{ _____}.$$

रेखाएँ $L_1 : 2x + y = 2$, $L_2 : x - 2y = -1$ और L_3 हैं, जहाँ L_3 वह रेखा है जो L_1 और L_2 के प्रतिच्छेदन बिन्दु से होकर गुजरती है। रेखा L बिन्दु $P(0, 1)$ से गुजरती है और रेखाएँ L_1 , L_2 तथा L_3 को क्रमशः बिन्दु A , B तथा C पर काटती है। यदि

रेखा L का L_1 और L_2 के मध्य अन्तःखण्ड मूल बिन्दु पर समकोण बनाता है तथा $\left(\frac{1}{PA}\right)^2 + \left(\frac{1}{PB}\right)^2 + PC^2 = 5$, L_3 और L का समीकरण ज्ञात करें _____.

Ans.

Sol.

$$\text{Let } L_3 \equiv (2x + y - 2) + \lambda (x - 2y + 1) = 0$$

$$L \equiv \frac{x-0}{\cos \theta} = \frac{y-1}{\sin \theta} = r$$

$$A \equiv (r_1 \cos \theta, r_1 \sin \theta + 1); B(r_2 \cos \theta, r_2 \sin \theta + 1) \text{ and } C(r_3 \cos \theta, r_3 \sin \theta + 1)$$

$$A \text{ satisfies } L_1 \Rightarrow \frac{1}{r_1} = 2 \cos \theta + \sin \theta; B \text{ satisfies } L_2 \Rightarrow \frac{1}{r_2} = \cos \theta - 2 \sin \theta$$

$$C \text{ satisfies } L_1 \Rightarrow \frac{1}{r_3} = \frac{1+\lambda}{(2+\lambda)\cos \theta + (1-2\lambda)\sin \theta}$$

Since intercept of line L between L_1 and L_2 subtends 90° at origin homogenizing L_1 and L_2 with $L \equiv y - mx - 1 = 0$,

$$\text{we get : } 2x^2 - 2y^2 - 3xy + 5y(y - mx) - 2(y - mx)^2 = 0$$

$$\text{Coefficient } x^2 + \text{coefficient } y^2 = 0$$

$$\Rightarrow 5 - 2(1 + m^2) = 0 \Rightarrow m = \pm \sqrt{\frac{3}{2}}$$

$$L \text{ is } y = \pm \sqrt{\frac{3}{2}}x + 1$$

$$\text{Also } \left(\frac{1}{PA}\right)^2 + \left(\frac{1}{PB}\right)^2 + \left(\frac{1}{PC}\right)^2 = 5$$

$$(2 \cos \theta + \sin \theta)^2 + (\cos \theta - 2 \sin \theta)^2 + \left(\frac{1+\lambda}{(2+\lambda)\cos \theta + (1-2\lambda)\sin \theta}\right)^2 = 5$$

$$\Rightarrow (1 + \lambda)^2 = 0 \Rightarrow \lambda = -1$$

$$\text{Equation of } L_3 \text{ is } x + 3y = 3$$

C01/4/157. Consider a curve $ax^2 + 2hxy + by^2 = 1$ and a point P not on the curve. A line drawn from the point P intersect the curve at points Q and R . If the product $PQ \cdot PR$ is independent of the slope of the line, then show that the curve is a circle _____.

मान लें कि एक वक्र $ax^2 + 2hxy + by^2 = 1$ है और बिन्दु P इस वक्र पर नहीं स्थित है। बिन्दु P से खींची गई एक रेखा इस वक्र

को बिन्दुओं Q और R पर काटती हैं। यदि $PQ \cdot PR$ का मान रेखा के ढाल पर निर्भर नहीं करता है, तो यह सिद्ध करें कि वक्र एक वृत्त है _____.

Ans.

Sol.

The given circle is $ax^2 + 2hxy + by^2 = 1$ (i)

Let the point P not lying on (i) be (x_1, y_1) , let θ be the inclination of line through P which intersects the given

curve at Q and R.

Then, equation of line through P is $\frac{x-x_1}{\cos\theta} = \frac{y-y_1}{\sin\theta} = r$

$$\Rightarrow x = x_1 + r \cos\theta, y = y_1 + r \sin\theta$$

For points Q and R, above point must lie on equation (i)

$$\Rightarrow a(x_1 + r \cos\theta)^2 + 2h(x_1 + r \cos\theta)(y_1 + r \sin\theta) + b(y_1 + r \sin\theta)^2 = 1$$

$$\Rightarrow (a \cos^2\theta + 2h \sin\theta \cos\theta + b \sin^2\theta)r^2 + 2(ax_1 \cos\theta + hx_1 \sin\theta + hy_1 \cos\theta + by_1 \sin\theta)r + (ax_1^2 + 2hx_1y_1 + by_1^2 - 1) = 0$$

It is quadratic in r, giving two values of r as PQ and PR.

$$\therefore PQ \cdot PR = \frac{ax_1^2 + 2hx_1y_1 + by_1^2 - 1}{a \cos^2\theta + 2h \sin\theta \cos\theta + b \sin^2\theta}$$

Here, $ax_1^2 + 2hx_1y_1 + by_1^2 - 1 \neq 0$, as (x_1, y_1) does not lie on equation (i).

Also, $a \cos^2\theta + 2h \sin\theta \cos\theta + b \sin^2\theta$

$$= a + \sin\theta \{2h \cos\theta + (b-a) \sin\theta\}$$

$$= a + \sin\theta \sqrt{4h^2 + (b-a)^2} \cdot (\cos\theta \sin\phi + \sin\theta \cos\phi)$$

$$\text{Where, } \tan\theta = \frac{b-a}{2h} = a + \sqrt{4h^2 + (b-a)^2} \sin\theta + \sin(\theta + \phi)$$

Which will be independent of θ , if $4h^2 + (b-a)^2 = 0 \Rightarrow h = 0$ and $b = a$

$$\therefore \text{Equation (i) reduces to } x^2 + y^2 = \frac{1}{a}$$

Which is a equation of circle

C02/4/158. C_1 and C_2 are two concentric circles, the radius of C_2 being twice that of C_1 . From a point P on C_2 , tangents PA and PB are drawn to C_1 . Prove that the centroid of the ΔPAB lies on C_1 . _____.

C_1 और C_2 दो संकेन्द्रीय वृत्त हैं, जिनमें C_2 की त्रिज्या C_1 की त्रिज्या से दोगुनी है। C_2 पर स्थिति एक बिन्दु P से C_1 पर स्पर्श रेखाएँ PA और PB खींची जाती हैं। सिद्ध करें कि त्रिभुज PAB का केन्द्रक C_1 पर स्थित है _____.

Ans.

Sol.

Let the coordinate of point P be $(2r \cos\theta, 2r \sin\theta)$

We have, $OA = r$, $OP = 2r$

Since ΔOAP is a right angled triangle.

$$\cos \phi = 1/2 \quad \Rightarrow \phi = \pi/3$$

$\therefore$ Coordinates of A are $\{r \cos(\theta - \pi/3), r \sin(\theta - \pi/3)\}$ and that of B are $[r \cos(\theta + \pi/3), r \sin(\theta + \pi/3)]$

If p, q is the centroid of ΔPAB , then

$$p = \frac{1}{3} \left[r \cos \left(\theta - \frac{\pi}{3} \right) + r \cos \left(\theta + \frac{\pi}{3} \right) + 2r \cos \theta \right] = \frac{1}{3} \left[r \left\{ \cos \left(\theta - \frac{\pi}{3} \right) + \cos \left(\theta + \frac{\pi}{3} \right) \right\} + 2r \cos \theta \right]$$

$$= \frac{1}{3} \left[r \left(2 \cos \frac{\theta - \frac{\pi}{3} + \theta + \frac{\pi}{3}}{2} \cdot \cos \frac{\theta - \frac{\pi}{3} - \theta - \frac{\pi}{3}}{2} \right) + 2r \cos \theta \right] = \frac{1}{3} \left[r \left\{ 2 \cos \theta \cos \frac{\pi}{3} \right\} + 2r \cos \theta \right]$$

$$= \frac{1}{3} [r \cdot \cos \theta + 2r \cos \theta] = r \cos \theta \text{ and}$$

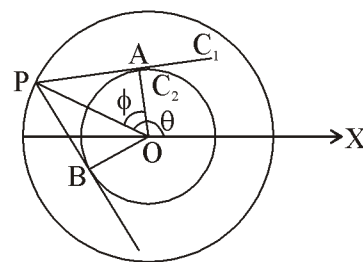
$$q = \frac{1}{3} \left[r \sin \left(\theta - \frac{\pi}{3} \right) + r \sin \left(\theta + \frac{\pi}{3} \right) + 2r \sin \theta \right]$$

$$= \frac{1}{3} \left[r \left\{ \sin \left(\theta - \frac{\pi}{3} \right) + \sin \left(\theta + \frac{\pi}{3} \right) \right\} + 2r \sin \theta \right]$$

$$= \frac{1}{3} \left[r \left(2 \sin \frac{\theta - \frac{\pi}{3} + \theta + \frac{\pi}{3}}{2} \cdot \cos \frac{\theta - \frac{\pi}{3} - \theta - \frac{\pi}{3}}{2} \right) + 2r \sin \theta \right]$$

$$= \frac{1}{3} \left[r \left(2 \sin \theta \cos \frac{\pi}{3} \right) + 2r \sin \theta \right] = \frac{1}{3} [r (\sin \theta) + 2r \sin \theta] = r \sin \theta$$

Now, $(p, q) = (r \cos \theta, r \sin \theta)$ lies on $x^2 + y^2 = r^2$ which is C_1 .



C02/4/159. Find the equations of the circles passing through $(-4, 3)$ and touching the lines $x + y = 2$ and $x - y = 2$ _____.

बिन्दु $(-4, 3)$ से गुजरने वाले तथा रेखाओं $x + y = 2$ और $x - y = 2$ को स्पर्श करने वाले वृत्तों के समीकरण होंगे _____.

Ans.

Sol.

Let the equation of the required circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \text{_____ (i)}$$

It passes through $(-4, 3)$.

$$\therefore 25 - 8g + 6f + c = 0 \quad \text{_____ (ii)}$$

Since, circle touches the line $x + y - 2 = 0$ and $x - y - 2 = 0$

$$\therefore \left| \frac{-g-f-2}{\sqrt{2}} \right| = \left| \frac{-g+f-2}{\sqrt{2}} \right| = \sqrt{g^2 + f^2 - c} \quad \text{_____ (iii)}$$

$$\text{Now, } \left| \frac{-g-f-2}{\sqrt{2}} \right| = \left| \frac{-g+f-2}{\sqrt{2}} \right| \Rightarrow -g-f-2 = \pm (-g+f-2)$$

$$\Rightarrow -g-f-2 = -g+f-2 \text{ or } -g-f-2 = g-f+2 \Rightarrow f=0 \text{ or } g=-2$$

Case I When $f=0$

$$\text{From equation (iii), we get : } \left| \frac{-g-2}{\sqrt{2}} \right| = \sqrt{g^2 - c} \Rightarrow (g+2)^2 = 2(g^2 - c)$$

$$\Rightarrow g^2 - 4g - 4 - 2c = 0 \quad \text{_____ (iv)}$$

On putting $f=0$ in equation (ii), we get :

$$25 - 8g + c = 0 \quad \text{_____ (v)}$$

Eliminating c between equation (iv) and (v), we get : $g^2 - 20g + 46 = 0$

$$\Rightarrow g = 10 \pm 3\sqrt{6} \text{ and } c = 55 \pm 24\sqrt{6}$$

On substituting the values of g , f and c in equation (i), we get : $x^2 + y^2 + 2(10 \pm 3\sqrt{6})x + (55 \pm 24\sqrt{6}) = 0$

Case II When $g = -2$

From equation (iii), we get :

$$\Rightarrow f^2 = 2(4 + f^2 - c) \Rightarrow f^2 - 2c + 8 = 0 \quad \text{..... (vi)}$$

On putting $g = -2$ in equation (ii), we get : $c = -6f - 41$

On substituting c in equation (vi), we get : $f^2 + 12f + 90 = 0$

This equation gives imaginary values off. Thus, there is no circle in this case.

Hence, the required equations of the circles are $x^2 + y^2 + 2(10 \pm 3\sqrt{6})x + (55 \pm 24\sqrt{6}) = 0$

C02/4/160. Two sides of a triangle are given by pair of straight lines $y^2 - 8xy - 7x^2 = 0$ and mid-point of third side is $(6/13, 1/13)$. Find the equation of the circum-circle of this triangle. _____.

एक त्रिभुज की दो भुजाएँ सरल रेखाओं के युग्म $y^2 - 8xy - 7x^2 = 0$ द्वारा दी गई हैं तथा $(6/13, 1/13)$ तीसरी भुजा का मध्य बिन्दु है। इस त्रिभुज के परिवृत्त का समीकरण ज्ञात कीजिए।

Ans. $13x^2 + 13y^2 - 17x - 13y = 0$

Sol. Given equation of pair of straight lines is $y^2 - 8xy - 7x^2 = 0$ (i)

Clearly, third side will be a chord of this curve with mid-point $(6/13, 1/13)$.

$$\text{Hence, equation of this third side is } y \cdot \frac{1}{13} - 4 \left(x \cdot \frac{1}{13} + y \cdot \frac{6}{13} \right) - 7 \cdot x \cdot \frac{6}{13} = \frac{1}{169} - \frac{48}{169} - \frac{252}{169}$$

$$\Rightarrow \frac{1+2a}{-7} = \frac{1+b}{1} = \frac{a+2b}{-8} \quad \Rightarrow 2x + y = 1$$

Let the equation of circle be $x^2 + y^2 + ax + by = 0$ [as circle will pass through point of intersection of pair of straight lines i.e. through origin).

Now, homogenizing this equation with the help of line $2x + y = 1$, we get : $x^2 + y^2 + (ax + by)(2x + y) = 0$

$$\Rightarrow (1 + 2a)x^2 + (1 + b)y^2 + (a + 2b)xy = 0 \quad \text{_____ (i)}$$

$$\text{This equation must be same as } y^2 - 8xy - 7x^2 = 0 \quad \text{_____ (ii)}$$

Comparing (i) and (ii)

$$\Rightarrow \frac{1+2a}{-7} = \frac{1+b}{1} = \frac{a+2b}{-8}$$

$$\Rightarrow 1 + 2a = -7 - 7b \quad \Rightarrow 7b + 2a = -8 \text{ and } -8 - 8b = a + 2b \quad \Rightarrow 10b + a = -8$$

$$\Rightarrow \frac{b}{8} = \frac{a}{80-56} = \frac{1}{7-20} \quad \Rightarrow a = \frac{24}{-13} \text{ and } b = \frac{8}{-13}$$

$$\therefore \text{Equation of circum-circle is } x^2 + y^2 - \frac{24}{13}x - \frac{8}{13}y = 0$$

C02/4/161. From any arbitrary point P on $y = \cos x$, tangents PA and PB are drawn to a circle which passes through the points (1, 0), (3, 0) and touches the circle $x^2 + y^2 - 2x - 8 = 0$ and have its centre in first quadrant. Find the locus of circumcentre of triangle PAB _____.

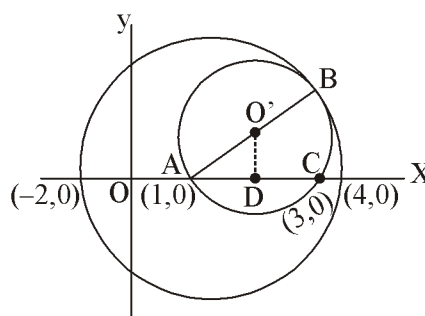
$y = \cos x$ पर किसी यादृच्छिक बिन्दु P से एक वृत्त पर स्पर्श रेखाएँ PA तथा PB खींची जाती हैं जो बिन्दुओं (1, 0), (3, 0) से गुजरती हैं तथा वृत्त $x^2 + y^2 - 2x - 8 = 0$ को स्पर्श करती हैं और इसका केन्द्र प्रथम चतुर्थांश में है। त्रिभुज PAB के परिकेन्द्र का बिन्दुपथ ज्ञात कीजिए।

Ans. $y = \frac{1}{2} \cos(2x - 2) + \frac{\sqrt{5}}{4}$

Sol. Here AB is the diameter of required circle and midpoint (O') of AB is centre of the required circle. Let O' = (h, k)
h = OD (here D is midpoint of AC)
 $\Rightarrow h = 2$

$$\text{and } k = O'D = \sqrt{O'A^2 - AD^2} = \sqrt{\frac{9}{4} - 1} = \frac{\sqrt{5}}{2}$$

Hence equation of the required circle is



$$(x-2)^2 + \left(y - \frac{\sqrt{5}}{2}\right)^2 = \frac{9}{4}$$

$$\Rightarrow x^2 + y^2 - 4x - \sqrt{5}y + 3 = 0 \quad \text{_____ (1)}$$

Let point P be $(x_0, \cos x_0)$. From P, tangent PA and PB are drawn to circle (1), circumcircle of triangle PAB also passes through centre of circle (1) i.e. O' and PO' is the diameter of the circumcircle of triangle PAB. Let midpoint of PO' is (x', y') .

$$\Rightarrow x' = \frac{x_0 + 2}{2} \quad \text{_____ (2)} \quad \text{and} \quad y' = \frac{\cos x_0 + \frac{\sqrt{5}}{2}}{2} \quad \text{_____ (3)}$$

From (2) and (3), on eliminating x_0 we get:

$$y' - \frac{\sqrt{5}}{2} = \cos(2x' - 2) \Rightarrow 2y - \frac{\sqrt{5}}{2} = \cos(2x - 2) \text{ is the required locus.}$$

- C02/4/162. A circle with radius a and centre at y-axis slides along it and a variable line through $(a, 0)$ cuts the circle at P and Q. Find the region in which point of intersection of tangent to the circle at P and Q will lie _____.

त्रिज्या a तथा y-अक्ष पर केन्द्र वाला एक वृत्त इसके अनुदिश घूमता रहता है तथा $(a, 0)$ से गुजरने वाली एक रेखा वृत्त को बिन्दु P तथा Q पर काटती है। वह क्षेत्र ज्ञात कीजिए जिसमें वृत्त पर P तथा Q पर स्पर्श रेखाओं का प्रतिच्छेद बिन्दु स्थित होगा _____.

Ans. $y^2 \geq 4a(x - a)$

Sol. Let the centre of circle be $(0, b)$

$$\text{Equation of circle } x^2 + (y - b)^2 = a^2$$

Let tangents at P and Q intersect at R (h, k) .

$$\text{Equation of chord of contact is } xh + yk - b(y + k) + b^2 - a^2 = 0 \text{ it passes through } (a, 0)$$

$$\Rightarrow b^2 - kb + ah - a^2 = 0 \text{ it is a quadratic in } b \text{ where } b \in \mathbb{R}$$

$$\Rightarrow k^2 - 4a(h - a) \geq 0$$

$$y^2 - 4a(x - a) \geq 0.$$

- V01/4/163. Let OA, OB, OC be coterminal edges of a cuboid. If ℓ, m, n be the shortest distances between the sides OA, OB, OC and their respective skew body diagonals to them, respectively, then find

$$\frac{\left(\frac{1}{\ell^2} + \frac{1}{m^2} + \frac{1}{n^2}\right)}{\left(\frac{1}{OA^2} + \frac{1}{OB^2} + \frac{1}{OC^2}\right)} \quad \text{_____}.$$

माना OA, OB, OC एक घनाभ की सहवर्ती भुजाएँ हैं। यदि ℓ, m, n भुजाओं OA, OB, OC और उनके तिरछे विकर्णों के

मध्य की सबसे कम दूरियाँ हैं तब $\frac{\left(\frac{1}{\ell^2} + \frac{1}{m^2} + \frac{1}{n^2}\right)}{\left(\frac{1}{OA^2} + \frac{1}{OB^2} + \frac{1}{OC^2}\right)}$ का मान ज्ञात कीजिए।

Ans. 2

Sol. Let $\ell = m = n = \frac{1}{\sqrt{2}}$

V01/4/164. Let OABC be a tetrahedron whose edges are of unit length. If $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \alpha(\vec{a} + \vec{b}) + \beta(\vec{a} \times \vec{b})$ then $(\alpha\beta)^2 = \frac{p}{q}$ where p and q are relatively prime to each other. Find the value of $\left[\frac{q}{2p}\right]$ where [.] denotes greatest integer function _____.

माना OABC एक एक चतुष्फलक है जिसकी भुजाओं की लम्बाई इकाई है। यदि $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ तथा $\overrightarrow{OC} = \alpha(\vec{a} + \vec{b}) + \beta(\vec{a} \times \vec{b})$ तब $(\alpha\beta)^2 = \frac{p}{q}$ है जहाँ p तथा q एक दूसरे के सापेक्ष अभाज्य हैं। $\left[\frac{q}{2p}\right]$ का मान ज्ञात कीजिए जहाँ [.] महत्तम पूर्णांक फलन को दर्शाता है।

Ans. 5

Sol. $\overrightarrow{OC}^2 = \alpha^2(\vec{a} + \vec{b})^2 + \beta^2(\vec{a} \times \vec{b})^2 + 2\alpha\beta[\vec{a} \cdot (\vec{a} \times \vec{b}) + \vec{b} \cdot (\vec{a} \times \vec{b})]$

$$\Rightarrow 1 = \alpha^2 \left(1 + 1 + 2 \cdot 1 \cdot 1 \cdot \frac{1}{2}\right) + \beta^2 \cdot 1 \cdot 1 \left(\frac{\sqrt{3}}{2}\right)^2 + 0 \dots\dots\dots(1)$$

$$\text{Also } \overrightarrow{OB} \cdot \overrightarrow{OC} = |\overrightarrow{OB}| \cdot |\overrightarrow{OC}| \cos \frac{\pi}{3}$$

$$\Rightarrow \alpha \cdot 1 \cdot 1 \cdot \frac{1}{2} + \alpha \cdot 1 = \frac{1}{2} \quad \Rightarrow \alpha = \frac{1}{3} \quad \dots\dots\dots(2)$$

$$\therefore \text{From (1) and (2) } \beta^2 = \frac{8}{9}$$

V01/4/165. Let $\vec{v}_0$ be a fixed vector and $\vec{v}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Then for $n \geq 0$ a sequence is defined

$$\vec{v}_{n+1} = \vec{v}_n + \left(\frac{1}{2}\right)^{n+1} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \vec{v}_0 \text{ then } \lim_{n \rightarrow \infty} \vec{v}_n = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}. \text{ Find } \frac{\alpha}{\beta} \text{ _____.}$$

माना $\vec{v}_0$ एक निश्चित सदिश है तथा $\vec{v}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ है। तब $n \geq 0$ के लिए एक अनुक्रम $\vec{v}_{n+1} = \vec{v}_n + \left(\frac{1}{2}\right)^{n+1} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \vec{v}_0$

परिभाषित है तब $\lim_{n \rightarrow \infty} \vec{v}_n = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ है। $\frac{\alpha}{\beta}$ का मान ज्ञात कीजिए।

Ans. 2

Sol.
$$\vec{v}_{n+1} - \vec{v}_n = \left(\begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix} \right)^{n+1} \vec{v}_0$$

$$\vec{v}_2 - \vec{v}_1 = \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix} \vec{v}_0 \quad \vec{v}_1 - \vec{v}_0 = \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix} \vec{v}_0$$

$$\vec{v}_3 - \vec{v}_2 = \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix}^3 \vec{v}_0$$

$$\vec{v}_n - \vec{v}_{n-1} = \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix}^n \vec{v}_0$$

adding all the equations

$$\vec{v}_n - \vec{v}_0 = (A + A^2 + A^3 + \dots + A^n) \vec{v}_0$$

$$\text{where } A = \begin{bmatrix} 0 & -1/2 \\ 1/2 & 0 \end{bmatrix} \Rightarrow \vec{v}_n = (I + A + A^2 + \dots) \vec{v}_0$$

V01/4/166. If $\vec{V}_1 = \hat{i} + \hat{j} + \hat{k}$; $\vec{V}_2 = a\hat{i} + b\hat{j} + c\hat{k}$ where $a, b, c \in \{-2, -1, 0, 1, 2\}$, If p is the number of non zero vectors $\vec{V}_2$ which are perpendicular to $\vec{V}_1$. Then sum of digits of p is _____.

यदि $\vec{V}_1 = \hat{i} + \hat{j} + \hat{k}$; $\vec{V}_2 = a\hat{i} + b\hat{j} + c\hat{k}$ जहाँ $a, b, c \in \{-2, -1, 0, 1, 2\}$ है तथा p अशून्य सदिश $\vec{V}_2$ की संख्या है जो $\vec{V}_1$ के लम्बवत् है तो p के अंकों का योग _____ होगा।

Ans. 9

Sol. We have

$$\vec{V}_1 \cdot \vec{V}_2 = a + b + c = 0$$

but $a, b, c \in \{-2, -1, 0, 1, 2\}$

now (i) if $a = 1, b = -1, c = 0$, number $= 3! = 6$

(ii) if $a = 2, b = -2, c = 0$, number $= 3! = 6$

(iii) if $a = 1, b = 1, c = -2$, number $= \frac{3!}{2!} = 3$

(iv) if $a = -1, b = -1, c = 2$, number $= \frac{3!}{2!} = 3$

$\therefore$ Total = 18

V01/4/167. Let P and Q are two points in xy-plane on the curve $y = x^7 - 2x^5 + 5x^3 + 8x + 5$ such that $\vec{OP} \cdot \hat{i} = 2$ and $\vec{OQ} \cdot \hat{i} = -2$ and the magnitude of $\vec{OP} + \vec{OQ} = 2M$, (where O is origin) then find the value of M_____.

माना P तथा Q, xy-समतल में वक्र $y = x^7 - 2x^5 + 5x^3 + 8x + 5$ पर दो बिन्दु इस प्रकार है कि $\vec{OP} \cdot \hat{i} = 2$ तथा $\vec{OQ} \cdot \hat{i} = -2$ तथा $|\vec{OP} + \vec{OQ}| = 2M$ का परिमाण है, (जहाँ O मूल बिन्दु है) तब M का मान _____ होगा।

Ans. 5

Sol. Let P and Q be (x_1, y_1) and (x_2, y_2)

$$\therefore \vec{OP} \cdot \hat{i} = x_1 = 2 \text{ and } \vec{OQ} \cdot \hat{i} = x_2 = -2$$

$$\text{Let } y = f(x) = x^7 - 2x^5 + 5x^3 + 8x + 5$$

$$\therefore y_1 = f(x_1) = f(2) \text{ and } y_2 = f(x_2) = f(-2)$$

$$\therefore \vec{OP} + \vec{OQ} = x_1\hat{i} + y_1\hat{j} + x_2\hat{i} + y_2\hat{j} = (f(2) + f(-2))\hat{j}$$

V01/4/168. If area of a triangular face BCD of a regular tetrahedron ABCD is $4\sqrt{3}$ sq. units, then find area of a triangle whose two sides are represented by vectors $\vec{AB}$ and $\vec{CD}$ _____.

चतुष्फलक ABCD के त्रिकोणीय फलक BCD का क्षेत्रफल $4\sqrt{3}$ वर्ग इकाई है तो त्रिभुज जिसकी भुजाएँ सदिश $\vec{AB}$ तथा $\vec{CD}$ से निरूपित करते हैं, का क्षेत्रफल _____ होगा।

Ans. 8

Sol. Let side be a

$$\frac{1}{2} a^2 \sin 60^\circ = \frac{4\sqrt{3}}{2}$$

$$a^2 = 16$$

$$\vec{AB} \cdot \vec{CD} = 0$$

$$\text{So area} = \frac{1}{2}a^2 \sin 90^\circ.$$

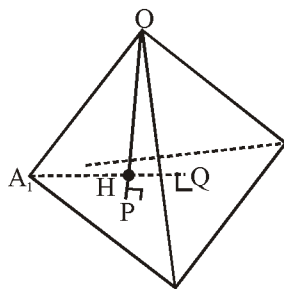
$$= \frac{a^2}{2} = 8$$

V01/4/169. Figure shows a tetrahedron $OA_1A_2A_3$. Altitudes OP and A_1Q are drawn from O and A_1 on opposite triangular faces. It is given that OP and A_1Q intersect at H . Let the lengths of OA_1 , A_2A_3 be l_1 and l_2 respectively. If shortest distance between OA_1 and A_2A_3 is d and volume of tetrahedron is V . Value of

$$\frac{dl_1l_2}{V} \text{ is } \underline{\hspace{2cm}}.$$

चतुष्फलक $OA_1A_2A_3$, चित्रानुसार दिया गया है। लम्ब OP तथा A_1Q , O तथा A_1 के विपरीत फलकों पर खींचे गये हैं। यदि दिया गया है कि OP तथा A_1Q , H पर प्रतिच्छेद करते हैं। माना OA_1 , A_2A_3 की लम्बाइयां क्रमशः l_1 तथा l_2 है। यदि OA_1

तथा A_2A_3 न्यूनतम दूरी d है तथा चतुष्फलक का आयतन V है तो $\frac{dl_1l_2}{V}$ का मान $\underline{\hspace{2cm}}$ होगा।



Ans. 6

Sol. $\vec{a}_1 \cdot (\vec{a}_2 - \vec{a}_3) = 0$

$$\vec{a}_4 \cdot (\vec{a}_2 - \vec{a}_3) = 0$$

$$(\vec{a}_1 - \vec{a}_4) \cdot (\vec{a}_2 - \vec{a}_3)$$

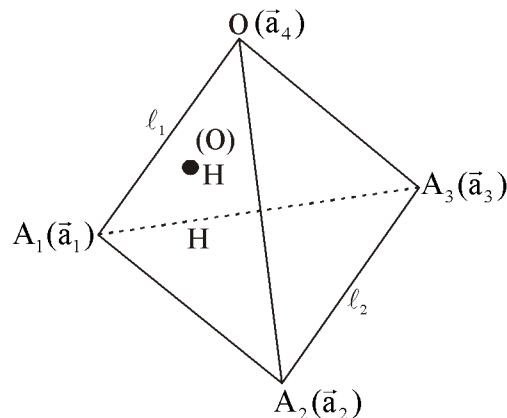
$$OA_1 \rightarrow \vec{r} = \vec{a}_4 + x(\vec{a}_1 - \vec{a}_4)$$

$$A_1A_3 \rightarrow \vec{r} = \vec{a}_2 + \mu(\vec{a}_2 - \vec{a}_3)$$

$$d = \frac{(\vec{a}_4 - \vec{a}_2) \cdot \{(\vec{a}_1 - \vec{a}_4) \times (\vec{a}_2 - \vec{a}_3)\}}{l_1l_2}$$

$$l_1l_2d = (\vec{a}_4 - \vec{a}_2) \cdot \{-\vec{a}_1 \times \vec{a}_2 + \vec{a}_1 \times \vec{a}_3 - \vec{a}_4 \times \vec{a}_2 + \vec{a}_4 \times \vec{a}_3\}$$

$$l_1l_2d = -[\vec{a}_1 \vec{a}_2 \vec{a}_4] + [\vec{a}_1 \vec{a}_3 \vec{a}_4]$$



V01/4/170. Let a_i , $i = 1, 2, 3, \dots, n$ denote integers in the domain of $f(x) = \sqrt{\log_{1/2} \left(\frac{4x-25}{x-21} \right)}$ where

$a_i < a_{i+1} \forall i \in \mathbb{N}$. If the line $\frac{2x-a_1}{4} = \frac{y+a_1}{a_2} = \frac{z-a_3}{a_5} = r$ meets the x - y , y - z and z - x planes at A , B and C respectively and volume of the tetrahedron $OABC$ is V , where ' O ' is origin and C' is the image of C in the x -axis then find the value of $\frac{9}{7}V$ _____.

माना a_i , $i = 1, 2, 3, \dots, n$, $f(x) = \sqrt{\log_{1/2} \left(\frac{4x-25}{x-21} \right)}$ के प्रान्त में पूर्णाकों को निरूपित करता है। जहाँ

$a_i < a_{i+1} \forall i \in \mathbb{N}$. यदि रेखा $\frac{2x-a_1}{4} = \frac{y+a_1}{a_2} = \frac{z-a_3}{a_5} = r$, x - y , y - z तथा z - x तल को क्रमशः A , B तथा C में मिलती

है तथा चतुष्फलक $OABC$ का आयतन V है, जहाँ ' O ' मूल बिन्दु है तथा C' x -अक्ष में C का प्रतिबिम्ब है, तो $\frac{9}{7}V$ का मान _____ होगा।

Ans.

4

Sol.

Domain of $f(x)$:

$$\log_{1/2} \left(\frac{4x-25}{x-21} \right) \geq 0$$

$$\Rightarrow 0 < \frac{4x-25}{x-21} \leq 1$$

$$x \in \left[\frac{4}{3}, \frac{25}{4} \right)$$

$\therefore$ Integers in the domain are 2, 3, ..., 6

$$a_1 = 2, a_2 = 3, \dots, a_5 = 6$$

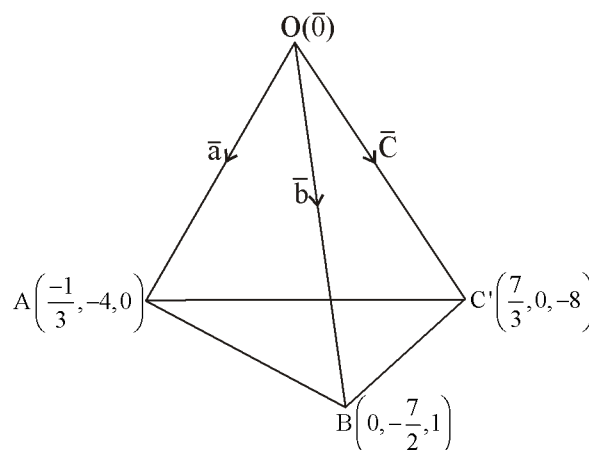
$$\text{Line : } \frac{x-1}{2} = \frac{y+2}{3} = \frac{z-4}{6} = r$$

$$\text{At } x\text{-}y \text{ plane } z = 0 \Rightarrow 6r + 4 = 0$$

$$\Rightarrow r = -\frac{2}{3}$$

$$A(2r+1, 3r-2, 0) \equiv \left(-\frac{1}{3}, -4, 0 \right)$$

$$\text{At } y\text{-}z \text{ plane } x = 0 \Rightarrow 2r + 1 = 0 \Rightarrow r = -\frac{1}{2}$$



$$B(0, 3r - 2, 6r + 4) \equiv \left(0, \frac{-7}{2}, 1\right)$$

$$\text{At } z\text{-x plane } y = 0 \Rightarrow 3r - 2 = 0 \Rightarrow r = \frac{2}{3}$$

$$C(2r + 1, 0, 6r + 4) \equiv \left(\frac{7}{3}, 0, 8\right) \Rightarrow C' \left(\frac{7}{3}, 0, -8\right)$$

$\therefore$ Volume of the tetrahedron OABC'

$$V = \frac{1}{6} [\vec{a}, \vec{b}, \vec{c}]$$

$$V = \frac{1}{6} \begin{vmatrix} -1/3 & -4 & 0 \\ 0 & -7/2 & 1 \\ 7/3 & 0 & -8 \end{vmatrix} = \left| -\frac{28}{9} \right|$$

V01/4/171. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three non-coplanar vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 3$ and angles between the vectors $\vec{a}$ and $\vec{b}$, $\vec{b}$ and $\vec{c}$, $\vec{c}$ and $\vec{a}$ are $\frac{\pi}{4}$, $\frac{\pi}{3}$ and $\frac{\pi}{2}$ respectively, then volume of parallelepiped formed with $\vec{a}$, $\vec{b}$ and $\vec{c}$ as coterminal edges is _____.

यदि $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ तीन असमतलीय सदिश इस प्रकार हैं कि $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 3$ तथा सदिशों $\vec{a}$ तथा $\vec{b}$, $\vec{b}$ तथा $\vec{c}$, $\vec{c}$ तथा $\vec{a}$ के मध्य कोण क्रमशः $\frac{\pi}{4}$, $\frac{\pi}{3}$ तथा $\frac{\pi}{2}$ हैं, तब सहवर्ती भुजाओं $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ से बनाए गए समांतर चतुर्भुज का आयतन होगा।

Ans. 3

Sol. Given : $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 3$

$$\vec{a} \cdot \vec{b} = ab \cos \frac{\pi}{4} = \sqrt{2}$$

$$\vec{a} \cdot \vec{b} = bc \cos \frac{\pi}{3} = 3$$

$$\vec{c} \cdot \vec{a} = ac \cos \frac{\pi}{2} = 0$$

$$\therefore [\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$[\vec{a} \quad \vec{b} \quad \vec{c}]^2 = \begin{vmatrix} 1 & \sqrt{2} & 0 \\ \sqrt{2} & 4 & 3 \\ 0 & 3 & 9 \end{vmatrix} = 9$$

$$\therefore V = [\vec{a} \quad \vec{b} \quad \vec{c}] = 3$$

V01/4/172. The vectors $(x^2 - 1)\hat{i} + 2(x^2 - 1)\hat{j} - 3(x^2 - 1)\hat{k}$, $(2x^2 - 1)\hat{i} + (2x^2 + 1)\hat{j} + x^2\hat{k}$, and $(3x^2 + 2)\hat{i} + (x^2 + 4)\hat{j} + (x^2 + 1)\hat{k}$ are non-coplanar. Then sum of real values that x can not take is _____.

सदिश $(x^2 - 1)\hat{i} + 2(x^2 - 1)\hat{j} - 3(x^2 - 1)\hat{k}$, $(2x^2 - 1)\hat{i} + (2x^2 + 1)\hat{j} + x^2\hat{k}$, तथा $(3x^2 + 2)\hat{i} + (x^2 + 4)\hat{j} + (x^2 + 1)\hat{k}$ असमतलीय है तो x के वास्तविक मानों का योग जो संभव नहीं है।

Ans. 0

Sol. The scalar triple product of the three given vectros is $\begin{vmatrix} x^2 - 1 & 2(x^2 - 1) & -3(x^2 - 1) \\ 2x^2 - 1 & 2x^2 + 1 & x^2 \\ 3x^2 + 2 & x^2 + 4 & x^2 + 1 \end{vmatrix}$

$$= (x^2 - 1) \begin{vmatrix} 1 & 2 & -3 \\ 2x^2 - 1 & 2x^2 + 1 & x^2 \\ 3x^2 + 2 & x^2 + 4 & x^2 + 1 \end{vmatrix} = (x^2 - 1) \begin{vmatrix} 1 & 2 & 0 \\ 2x^2 - 1 & 2x^2 + 1 & 5x^2 \\ 3x^2 + 2 & x^2 + 4 & 5x^2 + 7 \end{vmatrix} \text{ (Using } C_3 \rightarrow C_3 + C_2 + C_1)$$

$$= (x^2 - 1) \begin{vmatrix} 1 & 0 & 0 \\ 2x^2 - 1 & -2x^2 + 3 & 5x^2 \\ 3x^2 + 2 & -5x^2 & 5x^2 + 7 \end{vmatrix} \text{ (Using } C_2 \rightarrow C_2 - 2C_1)$$

$$= (x^2 - 1) (15x^4 + x^2 + 21)$$

Vector are non coplanar $\Rightarrow$ Scalar triple product $\neq 0$

$$\Rightarrow x^2 - 1 \neq 0 (\because 15x^4 + x^2 + 21 \neq 0) \Rightarrow x \neq \pm 1$$

V01/4/173. One vertex of a cube is $(0, 0, 0)$ and the sides are parallel to x , y and z - axis. Three planes $x = y$, $y = z$ and $z = x$ cut the cube in n parts. The value of n is _____.

एक घन का एक शीर्ष $(0, 0, 0)$ है तथा भुजाएँ x , y तथा z -अक्ष के साथ समानान्तर हैं। तीन तल $x = y$, $y = z$ तथा $z = x$ घन को n भागों में काटते हैं। n का मान होगा

Ans. 6

Sol. $x = y$ plane will cut the cube in two parts
 $y = z$ plane will cut in four parts
 $z = x$ will cut in 6 parts

V01/4/174. If $2\alpha - \beta + 3\gamma = 4$ & $\alpha + \beta + \gamma = 3$ ($\alpha, \beta, \gamma \in \mathbb{R}$), then minimum value of $\alpha^2 + \beta^2 + \gamma^2$ is _____.
 यदि $2\alpha - \beta + 3\gamma = 4$ तथा $\alpha + \beta + \gamma = 3$ ($\alpha, \beta, \gamma \in \mathbb{R}$) हो, तो $\alpha^2 + \beta^2 + \gamma^2$ का न्यूनतम मान _____ होगा।

Ans. 3

Sol. $2x - y + 3z = 4$
 $x + y + z = 3$

$$\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 3 \\ 1 & 1 & 1 \end{vmatrix} = -4\hat{i} + \hat{j} + 3\hat{k}$$

$$(\alpha, \beta, \gamma) \text{ lies on the line } \frac{x-1}{-4} = \frac{y-1}{1} = \frac{z-1}{3} = \lambda$$

$$\alpha = 1 - 4\lambda; \beta = \lambda + 1, \gamma = 3\lambda + 1$$

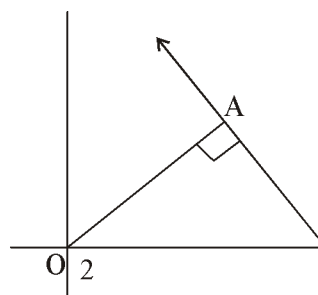
$$\text{for minimum distance } \vec{OA} \cdot \vec{b} = 0$$

$$16\lambda - 4 + \lambda + 1 + 9\lambda + 3 = 0$$

$$\lambda = 0$$

$$A \equiv (1, 1, 1)$$

$$\alpha^2 + \beta^2 + \gamma^2 = 3$$



V01/4/175. A plane P passes through the intersection of $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) + 3 = 0$ and $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 2$. If p divide the line segment joining $(3, 0, 2)$ and $(0, 3, -1)$ in the ratio 2 : 1 internally and the equation of plane is given by $\vec{r} \cdot (a\hat{i} - \hat{j} + b\hat{k}) = c$ where $a, b, c \in \mathbb{N}$ then $a + b + c$ is/are _____.

एक तल P, $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) + 3 = 0$ तथा $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 2$ के प्रतिच्छेदन से गुजरती है। यदि p, $(3, 0, 2)$ तथा

$(0, 3, -1)$ को जोड़ने वाली रेखा को 2 : 1 में अन्तःविभाजन करती है तथा तल का समीकरण $\vec{r} \cdot (a\hat{i} - \hat{j} + b\hat{k}) = c$ से दिया जाता है, जहाँ $a, b, c \in \mathbb{N}$ तो $a + b + c$ का मान _____ होगा।

Ans. 3

Sol. Required plane is $p_1 + \lambda p_2 = 0$ and passes through $(1, 2, 0)$

$$\Rightarrow p = (3\hat{i} - \hat{j} + 3\hat{k})a + b + c = 7$$

V01/4/176. If $\vec{a}, \vec{b}$ are two unit vectors and $\vec{c}$ is such that $\vec{c} = \vec{a} \times \vec{c} + \vec{b}$, then the maximum value of $[\vec{a} \ \vec{b} \ \vec{c}]$ is $\frac{p}{q}$ where p and q are co-prime natural numbers, then find the value of (p + q) _____.

यदि $\vec{a}, \vec{b}$ दो इकाई सदिश हैं तथा $\vec{c}$ इस प्रकार है कि $\vec{c} = \vec{a} \times \vec{c} + \vec{b}$, तब $[\vec{a} \ \vec{b} \ \vec{c}]$ का अधिकतम मान $\frac{p}{q}$ है। जहाँ p और q सहअभाज्य प्राकृत संख्याएँ हैं। तब (p + q) का मान _____ होगा।

Ans. 3

Sol. $\vec{c} = \vec{a} \times \vec{c} + \vec{b}$

$$\Rightarrow |\vec{c} - \vec{b}| = |\vec{a} \times \vec{c}| \Rightarrow c^2 + 1 - 2\vec{b} \cdot \vec{c} = c^2 \sin^2 \theta, \text{ where } \theta = \vec{a} \wedge \vec{c}$$

$$\Rightarrow 2\vec{b} \cdot \vec{c} = c^2 \cos^2 \theta + 1 \Rightarrow 2\vec{b} \cdot (\vec{a} \times \vec{c} + \vec{b}) = c^2 \cos^2 \theta + 1$$

$$\Rightarrow -2[\vec{a} \ \vec{b} \ \vec{c}] + 2 = c^2 \cos^2 \theta + 1$$

$$\Rightarrow 2[\vec{a} \ \vec{b} \ \vec{c}] = 1 - c^2 \cos^2 \theta \leq +1 \Rightarrow [\vec{a} \ \vec{b} \ \vec{c}] \leq 1/2$$

V01/4/177. If $[\vec{a} \ \vec{b} \ \vec{c}] = 2$, and $\vec{d}$ is unit vector then find the value of

$$|(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) + (\vec{b} \times \vec{c}) \times (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) \times (\vec{b} \times \vec{d})|. \text{ _____}$$

यदि $[\vec{a} \ \vec{b} \ \vec{c}] = 2$, तथा $\vec{d}$ इकाई सदिश है तब $|(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) + (\vec{b} \times \vec{c}) \times (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) \times (\vec{b} \times \vec{d})|$ का मान _____ होगा।

Ans. 4

Sol. Let $\vec{d} = \alpha \vec{a} + \beta \vec{b} + \gamma \vec{c}$

$$\text{Now } \vec{d} \cdot \vec{b} \times \vec{c} = 2\alpha$$

$$\vec{d} \cdot \vec{c} \times \vec{a} = 2\beta$$

$$\vec{d} \cdot \vec{a} \times \vec{b} = 2\gamma \quad \therefore [\vec{d} \ \vec{b} \ \vec{c}] \vec{a} + [\vec{d} \ \vec{c} \ \vec{a}] \vec{b} + [\vec{d} \ \vec{a} \ \vec{b}] \vec{c} = 2\vec{d}$$

$$\text{Now, } (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) + (\vec{b} \times \vec{c}) \times (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) \times (\vec{b} \times \vec{d})$$

$$= [\vec{a} \ \vec{b} \ \vec{d}] \vec{c} - 2\vec{d} + [\vec{b} \ \vec{c} \ \vec{d}] \vec{a} - 2\vec{d} + [\vec{c} \ \vec{a} \ \vec{d}] \vec{b} - 2\vec{d} = -4\vec{d}$$

V01/4/178. Let $\vec{a} = x^2\hat{i} - 3\hat{j} + (x+3)\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} - (x-3)\hat{k}$ be two vectors such that $|\vec{a}| = |\vec{b}|$. If $\vec{v}_1 = 2\vec{a} + 3\vec{b}$

and $\vec{v}_2 = 3\vec{a} - 2\vec{b}$ and $I = \int_1^3 (\vec{v}_1 \wedge \vec{v}_2) dx$, then find the value of $[I]$ _____.

[Note: $[k]$ denotes greatest integer function less than or equal to k .]

माना $\vec{a} = x^2\hat{i} - 3\hat{j} + (x+3)\hat{k}$ और $\vec{b} = \hat{i} + 3\hat{j} - (x-3)\hat{k}$ दो सदिश इस प्रकार हैं $|\vec{a}| = |\vec{b}|$. कि $\vec{v}_1 = 2\vec{a} + 3\vec{b}$ यदि

$\vec{v}_2 = 3\vec{a} - 2\vec{b}$ और $I = \int_1^3 (\vec{v}_1 \wedge \vec{v}_2) dx$, तब $[I]$ का मान _____ होगा।

[Note: जहाँ $[k]$ महत्तम पूर्णांकफलन है।]

Ans. 3

Sol. $\vec{a} \cdot \vec{b} = (x^2\hat{i} + 3\hat{j} + (x+3)\hat{k}) \cdot (\hat{i} + 3\hat{j} - (x-3)\hat{k})$

$$= x^2 - 9 - (x^2 - 9) = 0$$

$$\Rightarrow \vec{a} \perp \vec{b} \text{ and } |\vec{a}| \perp |\vec{b}|$$

$$\vec{v}_1 \cdot \vec{v}_2 = (2\vec{a} + 3\vec{b}) \cdot (3\vec{a} - 2\vec{b})$$

$$6a^2 - 6b^2 - 4\vec{a} \cdot \vec{b} + 9\vec{b} \cdot \vec{a} = 0$$

$$\therefore \vec{v}_1 \wedge \vec{v}_2 = \frac{\pi}{2}$$

$$\text{Now, } I = \int_1^3 (\vec{v}_1 \wedge \vec{v}_2) dx = \int_1^3 \frac{\pi}{2} dx = \frac{\pi}{2} (3-1) = \pi$$

$$\therefore [I] = 3$$

V01/4/179. $\vec{p} = 3^{(\sin x + \tan x)}\hat{i} + 3^{(\tan x - 1)}\hat{j} - 3^{\sin x}\hat{k}$ and $\vec{q} = 3^{(\sin x - 1)}\hat{i} + (3^{\sin x + \tan x})\hat{j} + (3^{\tan x})\hat{k}$ If sum of all values of x in

$[0, 50\pi]$ for which $\vec{p} \cdot \vec{q}$ is minimum is S , then find the value of $\left(\frac{S}{255\pi}\right)$ _____.

$\vec{p} = 3^{(\sin x + \tan x)}\hat{i} + 3^{(\tan x - 1)}\hat{j} - 3^{\sin x}\hat{k}$ और $\vec{q} = 3^{(\sin x - 1)}\hat{i} + (3^{\sin x + \tan x})\hat{j} + (3^{\tan x})\hat{k}$ यदि $[0, 50\pi]$ में x के सभी मानों

का योगफल जिसके लिए $\vec{p} \cdot \vec{q}$ न्यूनतम है, S है तब $\left(\frac{S}{255\pi}\right)$ का मान _____ होगा।

Ans. 5

Sol. $\vec{p} \cdot \vec{q} = 3^{2\sin x + \tan x - 1} + 3^{\sin x + 2\tan x - 1} - 3^{\sin x + \tan x}$

$$3\vec{p} \cdot \vec{q} = 3^{2\sin x + \tan x} + 3^{\sin x + 2\tan x} - 3 \cdot 3^{\sin x + \tan x}$$

$$3\vec{p} \cdot \vec{q} + 1 = \left(3^{\left(\frac{2\sin x + \tan x}{3} \right)} \right)^3 + \left(3^{\left(\frac{\sin x + 2\tan x}{3} \right)} \right)^3 + 1^3 - 3 \cdot \left(3^{\left(\frac{2\sin x + \tan x}{3} \right)} \right) + \left(3^{\left(\frac{\sin x + 2\tan x}{3} \right)} \right) \cdot 1$$

If a, b, c are positive then $a^3 + b^3 + c^3 - 3abc \geq 0$ and equality holds when $a = b = c$

यदि a, b, c धनात्मक हैं तब $a^3 + b^3 + c^3 - 3abc \geq 0$ तथा समता होगी जब $a = b = c$

$$\therefore \sin x = \tan x = 0 \quad \Rightarrow x = x\pi$$

$$S = 0 + \pi + 2\pi + \dots + 50\pi = \frac{\pi(50 \times 51)}{(2)}$$

$$\therefore \frac{S}{255\pi} = 5$$

V01/4/180. Four points with position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are coplanar such that

$(\sin \alpha)\vec{a} + (2 \sin 2\beta)\vec{b} + (3 \sin 3\gamma)\vec{c} - \vec{d} = 0$. Then, the least value of the expression

$\sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma$ is $\frac{p}{q}$, where p and q are coprime, then find (p + q). _____.

चार बिन्दुओं जिनके स्थिति सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ और $\vec{d}$ समतलीय प्रकार हैं कि $(\sin \alpha)\vec{a} + (2 \sin 2\beta)\vec{b} + (3 \sin 3\gamma)\vec{c} - \vec{d} = 0$.

तब व्यंजक $\sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma$ का न्यूनतम मान $\frac{p}{q}$ है। जहाँ P और q सहअभाज्य हैं तब (p + q) का मान _____ होगा है।

Ans. 15

Sol. $\sin \alpha + 2\sin 2\beta + 3\sin 3\gamma = 1$ _____ (i)

$$\text{also } |\sin \alpha + 2 \sin 2\beta + 3 \sin 3\gamma| \leq \sqrt{1+4+9} \sqrt{\sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma} \text{ as } |\vec{p} \cdot \vec{q}| \leq |\vec{p}| |\vec{q}|$$

$$\therefore \sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma \geq \frac{1}{14}$$

V01/4/181. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular vectors such that $|\vec{a}| = \sqrt{6}$, $|\vec{b}| = \sqrt{2}$, $|\vec{c}| = \sqrt{3}$ and a vector

$\vec{v} = (\vec{a} \times (\vec{a} \times \vec{b})) \times \vec{c} + (\vec{b} \times (\vec{b} \times \vec{c})) \times \vec{a} + (\vec{c} \times (\vec{c} \times \vec{a})) \times \vec{b}$ where magnitude of $\vec{v}$ is M then $\frac{M^2}{100}$ is _____.

यदि $\vec{a}, \vec{b}$ तथा $\vec{c}$ परस्पर लम्बवत् सदिश इस प्रकार हैं कि $|\vec{a}| = \sqrt{6}, |\vec{b}| = \sqrt{2}, |\vec{c}| = \sqrt{3}$ तथा एक सदिश

$\vec{v} = (\vec{a} \times (\vec{a} \times \vec{b})) \times \vec{c} + (\vec{b} \times (\vec{b} \times \vec{c})) \times \vec{a} + (\vec{c} \times (\vec{c} \times \vec{a})) \times \vec{b}$ है जहाँ $\vec{v}$ का परिमाण M है तो $\frac{M^2}{100}$ का मान है :

Ans. 3.96

Sol. $\therefore (\vec{a} \times (\vec{a} \times \vec{b})) \times \vec{c} = ((\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b}) \times \vec{c}$
 $= -|\vec{a}|^2 \vec{b} \times \vec{c} = -6(\vec{b} \times \vec{c})$

Similarly $(\vec{b} \times (\vec{b} \times \vec{c})) \times \vec{a} = -|\vec{b}|^2 \vec{c} \times \vec{a}$

$= -2(\vec{c} \times \vec{a})$ and $(\vec{c} \times (\vec{c} \times \vec{a})) \times \vec{b} = -|\vec{c}|^2 \vec{a} \times \vec{b} = -3(\vec{a} \times \vec{b})$

Now, $\vec{v} = -[3(\vec{a} \times \vec{b}) + 6(\vec{b} \times \vec{c}) + 2(\vec{c} \times \vec{a})]$

$\therefore \vec{a}, \vec{b}$ and $\vec{c}$ are mutually perpendicular

$\therefore \vec{a} \times \vec{b}, \vec{b} \times \vec{c}$ and $\vec{c} \times \vec{a}$ are also mutually perpendicular

$$\Rightarrow |\vec{v}| = \sqrt{9|\vec{a}|^2 |\vec{b}|^2 + 36|\vec{b}|^2 |\vec{c}|^2 + 4|\vec{c}|^2 |\vec{a}|^2} = 6\sqrt{11}$$

$$M = 6\sqrt{11}$$

$$M^2 = 36 \times 11$$

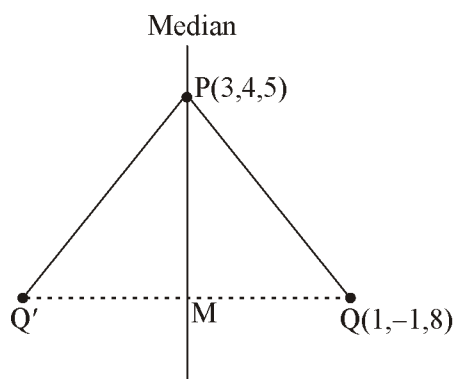
$$\frac{M^2}{100} = 3.96$$

V01/4/182. Consider a line passing through the point $P(3, 4, 5)$ & having direction ratios $(2, -2, 1)$. If image of point $Q(1, -1, 8)$ w.r.t. given line is Q' , then length of median of $\Delta PQQ'$ through vertex P is _____.

बिन्दु $P(3, 4, 5)$ से गुजरने वाली तथा दिक् अनुपातों $(2, -2, 1)$ वाली एक रेखा पर विचार कीजिए। यदि Q' , दी गई रेखा के सापेक्ष बिन्दु $Q(1, -1, 8)$ का प्रतिबिम्ब है, तो शीर्ष P से गुजरने वाली त्रिभुज PQQ' की माध्यिका की लम्बाई होगी :

Ans. 3.00

Sol.



$\Delta PQQ'$ is an isosceles triangle and $\overrightarrow{PM} = 2\hat{i} - 2\hat{j} + \hat{k}$

Now, $PM = |\overrightarrow{PQ}| \cos \theta$, $\overrightarrow{PQ} = 2\hat{i} + 5\hat{j} - 3\hat{k}$

$$PM = \frac{|\overrightarrow{PQ} \cdot \overrightarrow{PM}|}{|\overrightarrow{PM}|} = \frac{|4 - 10 - 3|}{3} = 3$$

V01/4/183. If lines $bx = 1 - cy$, $bz = 1$ & $bx = 2 + ay$, $bz = 3 + cy$ are perpendicular, where a, b, c are sides of ΔABC , then minimum value of $\cos B$ is _____.

यदि रेखाएँ $bx = 1 - cy$, $bz = 1$ तथा $bx = 2 + ay$, $bz = 3 + cy$ लम्बवत् हैं, जहाँ a, b, c त्रिभुज ΔABC की भुजाएँ हैं, तब $\cos B$ का न्यूनतम मान होगा :

Ans. 0.5

Sol. lines are $\frac{x - \frac{1}{b}}{c} = \frac{y}{-b} = \frac{z - \frac{1}{b}}{0}$ and $\frac{x - \frac{2}{b}}{a} = \frac{y}{b} = \frac{z - \frac{3}{b}}{c}$ are perpendicular

$$\Rightarrow ac - b^2 = 0 \Rightarrow b^2 = ac$$

$$\text{Now, } \cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{1}{2} \left(\frac{a}{c} + \frac{c}{a} \right) - \frac{1}{2} \geq \frac{1}{2} (2) - \frac{1}{2} \quad \left(\because x + \frac{1}{x} \geq 2, x > 0 \right)$$

$$\Rightarrow \text{minimum value of } \cos B = \frac{1}{2} \text{ exist when } a = c$$

V01/4/184. Let $|\vec{a}| = 4$, $|\vec{b}| = 1$ and $\vec{b} \times \vec{a} + \vec{a} = \vec{a}$. If projection of $\vec{a}$ along $\vec{b}$ is 3 then projection of $\vec{a}$ along $\vec{a}$ is _____.

माना $|\vec{a}| = 4, |\vec{b}| = 1$ तथा $\vec{b} \times \vec{\alpha} + \vec{a} = \vec{\alpha}$ हैं। यदि $\vec{\alpha}$ का $\vec{b}$ के सापेक्ष प्रक्षेप 3 है तो $\vec{\alpha}$ का $\vec{a}$ के सापेक्ष प्रक्षेप होगा _____.

Ans. 3.12 or 3.13

Sol. $\vec{b} \cdot (\vec{b} \times \vec{\alpha} + \vec{a}) = \vec{b} \cdot \vec{\alpha} \Rightarrow \vec{b} \cdot \vec{a} = \vec{b} \cdot \vec{\alpha}$ _____(1)

$\therefore$ proejction of $\vec{\alpha}$ on $\vec{b} = \vec{\alpha} \cdot \hat{b}$

$$\Rightarrow \vec{\alpha} \cdot \hat{b} = 3 \Rightarrow \vec{\alpha} \cdot \vec{b} = 3 \left(\because |\vec{b}| = 1 \right)$$

$$\therefore \vec{b} \cdot \vec{a} = 3 \text{ (from (1))}$$

$$\text{Now } \vec{b} \times (\vec{b} \times \vec{\alpha} + \vec{a}) = \vec{b} \times \vec{\alpha}$$

$$\Rightarrow (\vec{b} \cdot \vec{\alpha}) \vec{b} - b^2 \vec{\alpha} + \vec{b} \times \vec{a} = \vec{b} \times \vec{\alpha}$$

$$\Rightarrow 3\vec{b} - \vec{\alpha} + \vec{b} \times \vec{a} = \vec{b} \times \vec{\alpha}$$

$$\Rightarrow 3\vec{b} - \vec{\alpha} + \vec{b} \times \vec{a} = \vec{\alpha} - \vec{a}$$

$$\Rightarrow \vec{\alpha} = \frac{1}{2} (3\vec{b} + \vec{a} + \vec{b} \times \vec{a})$$

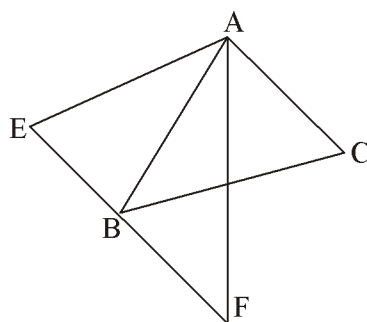
Projection of $\vec{\alpha}$ along $\vec{a}$

$$= \vec{\alpha} \cdot \hat{a} = \vec{\alpha} \cdot \frac{\vec{a}}{4} = \frac{1}{8} [3\vec{b} + \vec{a} + \vec{b} \times \vec{a}] \cdot \vec{a}$$

$$= \frac{1}{8} [3\vec{b} \cdot \vec{a} + |\vec{a}|^2] = \frac{1}{8} [3 \times 3 + 16] = \frac{25}{8} = 3.125$$

V01/4/185. If $\triangle ABC$ and $\triangle AEF$ are such that $EB = BF$, $AB = EF = 1$, $BC = 6$, $AC = \sqrt{33}$ and $\vec{AB} \cdot \vec{AE} + \vec{AC} \cdot \vec{AF} = 2$, then the value of $\vec{BC} \cdot \vec{BF}$ is (given figure)_____.

यदि $\triangle ABC$ तथा $\triangle AEF$ इस प्रकार हैं कि $EB = BF$, $AB = EF = 1$, $BC = 6$, $AC = \sqrt{33}$ तथा $\vec{AB} \cdot \vec{AE} + \vec{AC} \cdot \vec{AF} = 2$ हैं, तो $\vec{BC} \cdot \vec{BF}$ का मान होगा (दिया गया आरेख)_____.



Ans. 2

Sol. $\vec{AB} \cdot \vec{AE} + \vec{AC} \cdot \vec{AF} = 2$

$$\vec{AB} \cdot (\vec{AB} + \vec{BE}) + \vec{AC} \cdot (\vec{AB} + \vec{BF}) = 2$$

$$|\vec{AB}|^2 + \vec{AB} \cdot \vec{BE} + \vec{AC} \cdot \vec{AB} + \vec{AC} \cdot \vec{BF} = 2 \quad \text{---(1)}$$

$$\vec{AB} + \vec{CA} = -\vec{BC}$$

$$|\vec{AB} + \vec{CA}| = |-\vec{BC}|$$

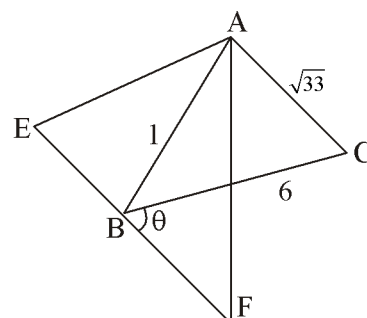
$$\Rightarrow 1 + 33 - 2\vec{AB} \cdot \vec{AC} = 36$$

$$\vec{AB} \cdot \vec{AC} = -1$$

$$(1) \Rightarrow \vec{AB} \cdot \vec{BE} + \vec{AC} \cdot \vec{BF} = 2 \quad (-\vec{AB} + \vec{AC}) \cdot \vec{BF} = 2 \quad [\because \vec{BE} = -\vec{BF}]$$

$$(\vec{BA} + \vec{AC}) \cdot \vec{BF} = 2$$

$$\vec{BC} \cdot \vec{BF} = 2$$



V01/4/186. In a tetrahedron LMNO edges ML, MN and MO are mutually perpendicular and length of altitudes drawn from O, L and N to opposite faces are 1, 2 and 3 units respectively, then length of altitude drawn from M to face LNO is _____.

एक चतुष्फलक LMNO की भुजाएँ ML, MN तथा MO परस्पर लम्बवत् हैं तथा O, L तथा N से विपरीत फलकों पर डाले गए लम्बों की लम्बाइयाँ क्रमशः 1, 2 तथा 3 इकाई हैं, तो M से फलक LNO पर डाले गए लम्ब की लम्बाई होगी _____.

Ans. 0.85 or 0.86

Sol. Let M being origin then one possible equation of face LNO will be

$$\Rightarrow x + \frac{y}{2} + \frac{z}{3} = 1$$

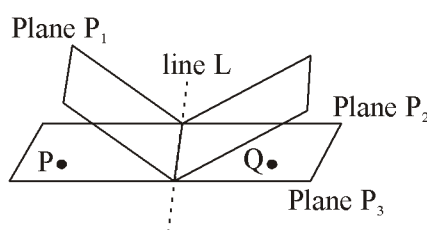
$$\therefore \text{required altitude} = \text{distance of LNO from M} = \frac{6}{7}$$

V01/4/187. In adjacent diagram equation of plane P_1 and P_2 are $x + y - 2z = 1$ and $\alpha x - y + z = 2$ respectively (where $\alpha \in \mathbb{Q}$). Coordinates of points P and Q are $(\alpha, 1, 3)$ and $\left(\frac{5}{3}, 1, \alpha\right)$ respectively on the plane P_3 .

if ℓ, m, n are direction cosines of line L then absolute value of $\frac{\sqrt{35}(\ell + m + n)}{\alpha}$ is _____.

दिए गए चित्र में समतलों P_1 तथा P_2 के समीकरण क्रमशः $x + y - 2z = 1$ तथा $\alpha x - y + z = 2$ हैं (जहाँ $\alpha \in \mathbb{Q}$)। समतल P_3 पर स्थित बिन्दुओं P तथा Q के निर्देशांक क्रमशः $(\alpha, 1, 3)$ तथा $\left(\frac{5}{3}, 1, \alpha\right)$ हैं। यदि ℓ, m, n रेखा L के दिक्कोज्याएँ हैं

तो $\frac{\sqrt{35}(\ell + m + n)}{\alpha}$ का निरपेक्ष मान होगा _____.



Ans. 4.50

Sol. equation of P_3 is $p_1 + \lambda p_2 = 0$
 $\Rightarrow (x + y - 2z - 1) + \lambda(\alpha x - y + z - 2) = 0$
 $\therefore$ P and Q lie on p_3

$$\Rightarrow (\alpha - 6) + \lambda(\alpha^2) = 0 \text{ and } \left(\frac{5}{3} - 2\alpha\right) + \lambda\left(\frac{5}{3}\alpha - 3 + \alpha\right) = 0$$

$$\Rightarrow 2\alpha^3 + \alpha^2 - 19\alpha + 18 = 0$$

$$\Rightarrow \alpha = 2 \text{ is only rational value.}$$

$\Rightarrow$ equation of planes are

$$P_1 : x + y - 2z = 1 \text{ and } P_2 : 2x - y + 2z = 2$$

$$\text{Vector along L is } \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -2 \\ 2 & -1 & 1 \end{vmatrix} = -\hat{i} - 5\hat{j} - 3\hat{k}$$

$\Rightarrow$ its direction cosines are $\pm \left(\frac{1}{\sqrt{35}}, \frac{5}{\sqrt{35}}, \frac{3}{\sqrt{35}} \right)$

V01/4/188. Let $\vec{r}$ and $\vec{s}$ be unit vectors. If $\vec{t}$ is a vector such that $\vec{t} + (\vec{t} \times \vec{r}) = \vec{s}$, the maximum volume of the parallelepiped formed by $\vec{r}, \vec{s}$ and $\vec{t}$ is $\frac{3}{2k}$, then k is equal to _____.

माना $\vec{r}$ तथा $\vec{s}$ इकाई सदिश हैं। यदि $\vec{t}$ एक सदिश इस प्रकार है कि $\vec{t} + (\vec{t} \times \vec{r}) = \vec{s}$, सदिशों $\vec{r}, \vec{s}$ तथा $\vec{t}$ से बने समांतर षट्फलक का अधिकतम आयतन $\frac{3}{2k}$ है, तो k का मान बराबर होगा :

Ans. 3

Sol. $\vec{t} + (\vec{t} \times \vec{r}) = \vec{s}$ _____ (1)

$\Rightarrow \vec{t} \cdot \vec{s} + [\vec{r} \vec{s} \vec{t}] = |\vec{s}|^2$ (taking dot product with $\vec{s}$)

$\Rightarrow [\vec{r} \vec{s} \vec{t}] = 1 - \vec{t} \cdot \vec{s}$

Also, $|\vec{t}|^2 = \vec{t} \cdot \vec{s}$ (taking dot product $\vec{t}$)

Squaring equation (1), we get $|\vec{t}|^2 + |\vec{t} \times \vec{r}|^2 + 2\vec{t} \cdot (\vec{t} \times \vec{r}) = |\vec{s}|^2$

$\Rightarrow |\vec{t}|^2 + |\vec{t}|^2 \sin^2 \theta + 0 = 1 \Rightarrow |\vec{t}|^2 = \frac{1}{1 + \sin^2 \theta} \geq \frac{1}{2}$

$\Rightarrow \vec{t} \cdot \vec{s} \geq \frac{1}{2} \Rightarrow 1 - \vec{t} \cdot \vec{s} \leq \frac{1}{2} \Rightarrow [\vec{r} \vec{s} \vec{t}] \leq \frac{1}{2}$

V01/4/189. The value of α for which $\hat{i} + 2\hat{j} + \hat{k}$, lies in the plane containing three points $\hat{i} + \hat{j} + \hat{k}, 2\hat{i} + 2\hat{j} + \hat{k}$ and $3\hat{i} - \hat{k}$ is _____.

α का मान जिसके लिए $\hat{i} + 2\hat{j} + \hat{k}$, तीन बिन्दुओं $\hat{i} + \hat{j} + \hat{k}, 2\hat{i} + 2\hat{j} + \hat{k}$ तथा $3\hat{i} - \hat{k}$ को समाहित करने वाले समतल में स्थित है, होगा :

Ans. 0.5

Sol. Let p.v. of given points be $A(\hat{i} + \hat{j} + \hat{k}), B(2\hat{i} + 2\hat{j} + \hat{k})$ and $C(3\hat{i} - \hat{k})$, so that two vectors in the plane may be $\overline{AB} = \hat{i} + \hat{j}$ and $\overline{AC} = 2\hat{i} - \hat{j} - 2\hat{k}$.

$$\text{Thus, } \begin{vmatrix} \alpha & 2 & 1 \\ 1 & 1 & 0 \\ 2 & -1 & -2 \end{vmatrix} = 0 \quad \Rightarrow -2\alpha - 2(-2) + (-1-2) = 0$$

$$\Rightarrow \alpha = \frac{1}{2}$$

V01/4/190. Let $\vec{x}, \vec{y}$ and $\vec{z}$ be unit vectors such that $\vec{x} + \vec{y} + \vec{z} = \vec{a}$, $\vec{x} \times (\vec{y} \times \vec{z}) = \vec{b}$, $(\vec{x} \times \vec{y}) \times \vec{z} = \vec{c}$, $\vec{a} \cdot \vec{x} = \frac{3}{2}$,

$\vec{a} \cdot \vec{y} = \frac{7}{4}$ and $|\vec{a}| = 2$. If $\vec{y} = -\lambda \vec{cm}$, then 3λ must be _____.

माना $\vec{x}, \vec{y}$ तथा $\vec{z}$ इकाई सदिश इस प्रकार हैं कि $\vec{x} + \vec{y} + \vec{z} = \vec{a}$, $\vec{x} \times (\vec{y} \times \vec{z}) = \vec{b}$, $(\vec{x} \times \vec{y}) \times \vec{z} = \vec{c}$, $\vec{a} \cdot \vec{x} = \frac{3}{2}$,

$\vec{a} \cdot \vec{y} = \frac{7}{4}$ तथा $|\vec{a}| = 2$ हैं। यदि $\vec{y} = -\lambda \vec{cm}$, तो 3λ का मान होगा :

Ans. 12

Sol. In first relation taking dot product with $\vec{a}$, we get $\vec{a} \cdot \vec{z} = \frac{3}{4}$

Now taking dot product of first relation with $\vec{x}$ and $\vec{y}$ and $\vec{z}$ in succession, we get

$$\vec{x} \cdot \vec{y} + \vec{x} \cdot \vec{z} = \frac{1}{2} \quad \text{_____ (1)}$$

$$\vec{x} \cdot \vec{y} + \vec{y} \cdot \vec{z} = \frac{3}{4} \quad \text{_____ (2)}$$

$$\vec{x} \cdot \vec{z} + \vec{y} \cdot \vec{z} = -\frac{1}{4} \quad \text{_____ (3) } \left(\because \vec{a} \cdot \vec{x} = \frac{3}{2}, \vec{a} \cdot \vec{y} = \frac{7}{4}, \vec{a} \cdot \vec{z} = \frac{3}{4} \right)$$

On solving (1), (2) and (3), we get

$$\vec{y} \cdot \vec{z} = 0, \vec{z} \cdot \vec{x} = -\frac{1}{4}, \vec{x} \cdot \vec{y} = \frac{3}{4} \quad \text{_____ (4)}$$

Now open the given relation

$$(\vec{x} \times \vec{y}) \times \vec{z} = \vec{c} \text{ and use (4)}$$

$$(\vec{x} \cdot \vec{z})\vec{y} - (\vec{y} \cdot \vec{z})\vec{x} = \vec{c}$$

$$\text{we get } \vec{y} = -4\vec{c} \Rightarrow \lambda = 4 \Rightarrow \lambda = 4 \Rightarrow 3\lambda = 12$$

V01/4/191. A ray light is coming along the line $\frac{x-2}{3} = \frac{y-1}{4} = \frac{z-6}{5}$ and strikes the plane mirror kept along the plane through the points (2, 1, 0), (5, 0, 1) and (4, 1, 1). Find the equation of reflected ray _____.

एक प्रकाश किरण रेखा $\frac{x-2}{3} = \frac{y-1}{4} = \frac{z-6}{5}$ के अनुदिश गमन करती है तथा बिन्दुओं (2, 1, 0), (5, 0, 1) तथा (4, 1, 1) से गुजरने वाले समतल दर्पण से परावर्तित होती है। परावर्तित किरण का समीकरण ज्ञात कीजिए।

Ans. $\frac{x+10}{4} = \frac{y+15}{5} = \frac{z+14}{3}$

Sol. Equation of plane passing through (2, 1, 0), (5, 0, 1) and (4, 1, 1) is $\begin{vmatrix} x-2 & y-1 & z-0 \\ 5-2 & 0-1 & 1-0 \\ 4-2 & 1-1 & 1-0 \end{vmatrix} = 0$

$$\Rightarrow (x-2)(-1) - (y-1)(3-2) + z(0+2) = 0$$

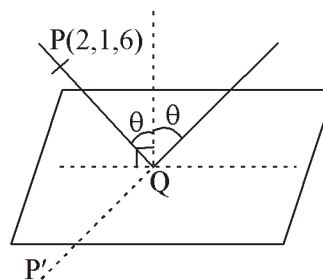
or, $x + y - 2z = 3$ _____ (i)

A point on the line $\frac{x-2}{3} = \frac{y-1}{4} = \frac{z-6}{5}$ is (2, 1, 6).

Now the reflection point of this point (2, 1, 6) w. r. t mirror kept along the plane $x + y - 2z = 3$ is

$$\frac{\alpha' - 2}{1} = \frac{\beta' - 1}{1} = \frac{\gamma' - 6}{-2} = \frac{-2(2+1-2 \times 6-3)}{1^2 + 1^2 + (-2)^2}$$

$$\Rightarrow \alpha' = 2 + 4 = 6, \beta' = 1 + 4 = 5 \text{ and } \gamma' = -8 + 6 = -2$$



Also the line $\frac{x-2}{3} = \frac{y-1}{4} = \frac{z-6}{5} = r$,

Let meets the plane

$$x + y - 2z = 3 \text{ at } (3r + 2, 4r + 1, 5r + 6)$$

$$\Rightarrow 3r + 2 + 4r + 1 - 2(5r + 6) = 3$$

$$\Rightarrow -3r = 12 \Rightarrow r = -4$$

$$\therefore \text{ point } Q = (-10, -15, -14)$$

$\therefore$ Equation of reflected ray is line joining p' and Q.

$$\Rightarrow \frac{x+10}{16} = \frac{y+15}{20} = \frac{z+14}{12}$$

$$\therefore \text{Reflected line } \frac{x+10}{4} = \frac{y+15}{5} = \frac{z+14}{3}$$

V01/4/192. $\hat{w}$ is a unit vector perpendicular to the plane of two other unit vectors $\hat{u}$ and $\hat{v}$. Three unit vectors $\hat{x}, \hat{y}$ and $\hat{z}$ are along the angle bisectors of $\hat{u}, \hat{v}; \hat{v}, \hat{w}$ and $\hat{v}, \hat{w}$ respectively. If $[\hat{x} \times \hat{y} \hat{y} \times \hat{z} \hat{z} \times \hat{x}] = \frac{1}{2}$ then find the angle between $\hat{u}$ and $\hat{v}$. _____.

$\hat{w}$ इकाई सदिश है जो दो अन्य इकाई सदिशों $\hat{u}$ तथा $\hat{v}$ वाले अन्य समतल के लम्बवत् है। तीन इकाई सदिश $\hat{x}, \hat{y}$ तथा $\hat{z}$ सदिशों $\hat{u}, \hat{v}; \hat{v}, \hat{w}$ तथा $\hat{v}, \hat{w}$ के कोण समद्विभाजकों के अनुदिश है। यदि $[\hat{x} \times \hat{y} \hat{y} \times \hat{z} \hat{z} \times \hat{x}] = \frac{1}{2}$ है तो $\hat{u}$ तथा $\hat{v}$ के मध्य कोण होगा :

Ans. $\pi/2$

Sol. Let angle between $\hat{u}$ and $\hat{v}$ be α then

$$\hat{x} = \frac{\hat{u} + \hat{v}}{2 \cos \frac{\alpha}{2}}, \hat{y} = \frac{\hat{v} + \hat{w}}{\sqrt{2}} \text{ and } \hat{z} = \frac{\hat{u} + \hat{w}}{\sqrt{2}}$$

$$\Rightarrow [\hat{x} \hat{y} \hat{z}] = \frac{1}{4 \cos \frac{\alpha}{2}} [\hat{u} + \hat{v} \quad \hat{v} + \hat{w} \hat{u} + \hat{w}] = \frac{2}{4 \cos \frac{\alpha}{2}} [\hat{u} \hat{v} \hat{w}]$$

$$\text{Also } [\hat{x} \times \hat{y} \hat{y} \times \hat{z} \hat{z} \times \hat{x}] = [\hat{x} \hat{y} \hat{z}]^2 = \left(\frac{[\hat{u} \hat{v} \hat{w}]}{2 \cos \alpha / 2} \right)^2 = \left(\frac{\sin \alpha}{2 \cos \alpha / 2} \right)^2 = \frac{1}{2}$$

$$\Rightarrow \alpha = \frac{\pi}{2}$$

V01/4/193. The lengths of two opposite edges of a tetrahedron are a, b; the shortest distance between these edges is d, and the angle between them is θ . Prove using vectors that the volume of the tetrahedron is $\frac{abd \sin \theta}{6}$ _____.

एक चतुष्फलक की दो विपरीत भुजाओं की लम्बाईयाँ a, b हैं; इन भुजाओं के मध्य न्यूनतम दूरी d है, तथा इनके मध्य कोण θ है। सदिश का प्रयोग कर सिद्ध कीजिए कि चतुष्फलक का आयतन $\frac{abd \sin \theta}{6}$ होगा।

Ans.

Sol. In the tetrahedron OABC, take O as the initial point and let the position vectors of A, B and C be $\vec{a}$, $\vec{k}$ and $\vec{c}$ respectively. the volume of the tetrahedron is equal to $\frac{1}{6} \vec{a} \cdot (\vec{k} \times \vec{c})$.

Also $\vec{BC} = \vec{c} - \vec{k}$ so that

Volume of tetrahedron i.e.

$$V = \frac{1}{6} \vec{a} \cdot (\vec{k} \times (\vec{k} + \vec{BC})) = \frac{1}{6} \vec{a} \cdot (\vec{k} \times \vec{BC}) = \frac{1}{6} \vec{k} \cdot (\vec{BC} \times \vec{a}) = \frac{1}{6} \vec{k} \cdot |\vec{BC}| |\vec{a}| \sin \theta \hat{n}, \text{ where } \hat{n} \text{ is the unit vector along PN, the line perpendicular to both OA and BC. Also } |\vec{BC}| = b.$$

$$\text{Here } V = \frac{1}{6} ab \sin \theta \vec{k} \cdot \hat{n} = \frac{1}{6} ab \sin \theta [\text{projection of OB on PN}]$$

$$= \frac{1}{6} ab \sin \theta [\text{perpendicular distance between OA and BC}] = \frac{1}{6} ab \sin \theta \cdot d = \frac{1}{6} abd \sin \theta.$$

V01/4/194. The position vectors of the vertices A, B and C of a triangle are $\hat{i} + \hat{j}$, $\hat{j} + \hat{k}$ and $\hat{i} + \hat{k}$ respectively. Find a unit vector $\hat{r}$ lying in the plane of $\triangle ABC$ and perpendicular to IA, where I is the incentre of the triangle _____.

एक त्रिभुज के शीर्षों A, B तथा C के स्थिति सदिश क्रमशः $\hat{i} + \hat{j}$, $\hat{j} + \hat{k}$ तथा $\hat{i} + \hat{k}$ हैं। त्रिभुज $\triangle ABC$ के तल में स्थित एक इकाई सदिश $\hat{r}$ ज्ञात कीजिए जो IA के लम्बवत् है, जहाँ I त्रिभुज का अन्तःकेन्द्र है।

Ans.

Sol. Since $|AB| = |BC| = |CA|$, incentre is same as circumcentre, and hence IA is perpendicular to BC
 $\Rightarrow \vec{r}$ is parallel to BC $\Rightarrow \vec{r} = \lambda(\hat{i} - \hat{j})$.

$$\text{Hence unit vector } \vec{r} = \pm \frac{1}{2}(\hat{i} - \hat{j})$$

MATCH THE COLUMN TYPE QUESTION(S)

V01/5/195. Match the following column :

निम्न कॉलम का सुमेलित कीजिए :

Column-I

Column-II

(a) Vectors $\vec{a}, \vec{b}, \vec{c}$ are of same magnitude when taken pairwise

(p) (6, 16, 20)

they makes equal angle if $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{i} + \hat{k}$. Then $\vec{c}$ is

- (b) If vector $\vec{r}$ satisfies $\vec{r} \times (2\hat{i} + \hat{j} + \hat{k}) = \hat{j} - \hat{k}$, then $\vec{r}$ equals (q) (0, 1, 1)
- (c) If plane passes through (3, 4, 5) and meets coordinate axes in A, B, C. Then point of intersection of plane through A, B, C parallel to coordinate planes can be (r) (3, 1, -1)
- (d) $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = 2\hat{i} - \hat{k}$, then point of intersection of line (s) $\left(\frac{4}{3}, -\frac{1}{3}, -\frac{1}{3}\right)$

$$\vec{r} \times \vec{a} = \vec{b} \times \vec{a} \text{ and } \vec{r} \times \vec{b} = \vec{a} \times \vec{b} \text{ is}$$

स्तम्भ-I

(t) (3, 2, 2)

स्तम्भ-II

- (a) सदिश $\vec{a}, \vec{b}, \vec{c}$ समान मापाक के हैं तथा जब युग्मवार लिये जाते हैं, (p) (6, 16, 20)

तब समान कोण बनाते हैं। यदि $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{i} + \hat{k}$ हो, तो $\vec{c}$ होगा

- (b) यदि सदिश $\vec{r}$, $\vec{r} \times (2\hat{i} + \hat{j} + \hat{k}) = \hat{j} - \hat{k}$ को सन्तुष्ट करता हो, तो (q) (0, 1, 1)

$\vec{r}$ होगा

- (c) यदि समतल बिन्दु (3, 4, 5) से गुजरता है तथा निर्देशी अक्षों को (r) (3, 1, -1)

A, B तथा C पर मिलता है। तब निर्देशी अक्षों के समान्तर

A, B, C से गुजरने वाले समतल के प्रतिच्छेद बिन्दु हो सकता है

- (d) यदि $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = 2\hat{i} - \hat{k}$ हो, तो रेखा $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ तथा (s) $\left(\frac{4}{3}, -\frac{1}{3}, -\frac{1}{3}\right)$

$\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ का प्रतिच्छेद बिन्दु होगा

(t) (3, 2, 2)

Choose the correct option :

सही विकल्प का चयन कीजिए :

(A) $a \rightarrow t$; $b \rightarrow q, s$; $c \rightarrow p$; $d \rightarrow r$

(B) $a \rightarrow t$; $b \rightarrow p$; $c \rightarrow q, s$; $d \rightarrow r$

(C) $a \rightarrow q, s$; $b \rightarrow t$; $c \rightarrow p$; $d \rightarrow r$

(D) $a \rightarrow t$; $b \rightarrow p$; $c \rightarrow r$; $d \rightarrow q, s$

Ans. C

Sol. (A) Let $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$$c_1^2 + c_2^2 + c_3^2 = 2,$$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{1}{2} = \frac{\vec{a} \cdot \vec{c}}{|\vec{a}| |\vec{c}|} = \frac{c_1 + c_2}{2} \Rightarrow c_1 + c_2 = 1$$

$$c_2 = 1 - c_1$$

$$\text{similarly } c_3 = 1 - c_1$$

$$\Rightarrow c_1^2 + (1 - c_1)^2 + (1 - c_1)^2 = 2 \Rightarrow c_1 = 0, \frac{4}{3}$$

(B) Let $\vec{r} = r_1\hat{i} + r_2\hat{j} + r_3\hat{k}$

$$\vec{r} \times (2\hat{i} + \hat{j} + \hat{k}) = \hat{j} - \hat{k}$$

$$\Rightarrow (r_2 - r_3)\hat{i} - \hat{j}(r_1 - 2r_3) + \hat{k}(r_1 - 2r_2) = \hat{j} - \hat{k}$$

comparing corresponding components

$$r_2 = r_3, r_1 - 2r_3 = -1, r_1 - 2r_2 = -1$$

$$\Rightarrow r_1 - 2r_2 = -1 \text{ There are infinite solvs.}$$

$$r_1 = \lambda, r_1 = -1 + 2r_2, r_3 = \lambda$$

$$\vec{r} = -1 + \lambda(2\hat{i} + \hat{j} + \hat{k}) \text{ if } \lambda = 2$$

$$\Rightarrow \vec{r} = 3\hat{i} + 2\hat{j} + 2\hat{k}$$

(C) Let equation of plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

$$\frac{3}{a} + \frac{4}{b} + \frac{5}{c} = 1 \Rightarrow A(a, 0, 0), B(0, b, 0), C(0, 0, c)$$

equation of plane through A and parallel to yz plane is $x = a$

equation of plane through B and parallel to zx plane is $y = b$

equation of plane through C and parallel to xy plane is (a, b, c)

But $a = 6, b = 16, c = 20$ satisfy

$\Rightarrow D$ can be correct answer

(D) $(\vec{r} - \vec{a}) \times \vec{b} = 0$

$$\Rightarrow \vec{r} = \vec{a} + t\vec{b}$$

$$= (\hat{i} + \hat{j}) + t(2\hat{i} - \hat{k})$$

$$= (1 + 2t)\hat{i} + \hat{j} - t\hat{k}$$

similarly second lines

$$\vec{r} = \vec{b} + s\vec{a}$$

$$\vec{r} = (2 + s)\hat{i} + s\hat{j} - \hat{k}$$

if two lines intersect

$$1 + 2t = 2s$$

$$1 = s$$

$$t = 1$$

$\Rightarrow$ Point of intersection is $(3, 1, -1)$

V01/5/196. Match the following column :

Column-I

(a) If the angle between the plane $x - 3y + 2z = 1$ and the

line $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-1}{-3}$ is θ , then the value of $|\operatorname{cosec}\theta|$ is

(b) If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar and $[\vec{a} \vec{b} \vec{c}] = \frac{4}{7}$,

Column-II

(p) 2

(q) 8

then $[2\vec{a} - \vec{b} \quad 2\vec{b} - \vec{c} \quad 2\vec{c} - \vec{a}]$ is

(c) Let $\vec{r} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$, where $\vec{a}, \vec{b}, \vec{c}$ (r) 4

are three non-coplanar vectors. If $\vec{r}$ is perpendicular

to $(\vec{a} + \vec{b} + \vec{c})$ and the minimum value of $(x^2 + y^2)$ is $\frac{k\pi^2}{4}$,

then 'k' is

(d) If image of the point (1,0,1) in the plane $x - y - z = 1$ (s) 5

is (a,b,c) then $3(a - b + c)$ is equal to

(t) an odd integer

स्तम्भ-I

स्तम्भ-II

(a) यदि समतल $x - 3y + 2z = 1$ तथा रेखा $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-1}{-3}$ (p) 2

के मध्य कोण θ हो, तो $|\operatorname{cosec} \theta|$ का मान होगा

(b) यदि $\vec{a}, \vec{b}, \vec{c}$ असमतलीय तथा $[\vec{a} \vec{b} \vec{c}] = \frac{4}{7}$ हो, (q) 8

तो $[2\vec{a} - \vec{b} \quad 2\vec{b} - \vec{c} \quad 2\vec{c} - \vec{a}]$ का मान होगा

(c) माना $\vec{r} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$ है, जहाँ $\vec{a}, \vec{b}, \vec{c}$ (r) 4

तीन असमतलीय सदिश हैं। यदि $\vec{r}, (\vec{a} + \vec{b} + \vec{c})$ के लबन्ध है तथा

$(x^2 + y^2)$ का न्यूनतम मान $\frac{k\pi^2}{4}$ हो, तो 'k' का मान होगा

(d) यदि बिन्दु (1,0,1) का, समतल $x - y - z = 1$ में प्रतिबिम्ब (a,b,c) है, (s) 5

तो $3(a - b + c)$ का मान होगा

(t) एक विषम पूर्णांक

Choose the correct option :

सही विकल्प का चयन कीजिए :

(A) (a) $\rightarrow$ (p); (b) $\rightarrow$ (r); (c) $\rightarrow$ (s,t); (d) $\rightarrow$ (q)

(B) (a) $\rightarrow$ (q); (b) $\rightarrow$ (s); (c) $\rightarrow$ (p,r); (d) $\rightarrow$ (p)

(C) (a) $\rightarrow$ (r); (b) $\rightarrow$ (t); (c) $\rightarrow$ (p,q); (d) $\rightarrow$ (r)

(D) (a) $\rightarrow$ (s); (b) $\rightarrow$ (p); (c) $\rightarrow$ (q,t); (d) $\rightarrow$ (p)

Ans. A

Sol. (A) Let angle between plane and line is θ .

$$\text{Also } |\sin \theta| = \left| \frac{(\hat{i} - 3\hat{j} + 2\hat{k}) \cdot (2\hat{i} + \hat{j} - 3\hat{k})}{14} \right| = \left| \frac{2 - 3 - 6}{14} \right| = \frac{1}{2}$$

$$\therefore |\operatorname{cosec} \theta| = 2$$

$$(B) [\vec{a} \vec{b} \vec{c}] = \frac{4}{7}$$

$$\begin{aligned}
 & [2\vec{a} - \vec{b} \quad 2\vec{b} - \vec{c} \quad 2\vec{c} - \vec{a}] \\
 & = \{(2\vec{a} - \vec{b}) \times (2\vec{b} - \vec{c})\} \cdot (2\vec{c} - \vec{a}) \\
 & = \{4(\vec{a} \times \vec{b}) - 2\vec{a} \times \vec{c} + \vec{b} \times \vec{c}\} \cdot (2\vec{c} - \vec{a}) \\
 & = 8[\vec{a} \vec{b} \vec{c}] - [\vec{b} \vec{c} \vec{a}] = 7[\vec{a} \vec{b} \vec{c}] \\
 & = 7 \times \frac{4}{7} = 4
 \end{aligned}$$

$$(C) \quad \vec{r} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$$

$$\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c}) = 0$$

$$\Rightarrow [\vec{a} \vec{b} \vec{c}] (\sin x + \cos y + 2) = 0$$

$$\Rightarrow \text{but } [\vec{a} \vec{b} \vec{c}] \neq 0$$

$$\Rightarrow \sin x + \cos y + 2 = 0$$

$$\Rightarrow \sin x = -1, \cos y = -1.$$

$\therefore$ For $x^2 + y^2$ to be minimum,

$$x = -\frac{\pi}{2}, y = \pi$$

$$\therefore x^2 + y^2 = \frac{\pi^2}{4} + \pi^2 = \frac{5\pi^2}{4}.$$

(D) Let the image of point A(1,0,1) in plane

$$x - y - z = 1$$

is A'(a,b,c)

D.R. of AA' are (a - 1, b, c - 1)

D.R. of normal are (1, -1, -1)

$$\Rightarrow \text{Let } \frac{a-1}{1} = \frac{b}{-1} = \frac{c-1}{-1} = \lambda \text{ (say)}$$

$$\Rightarrow a = \lambda + 1, b = -\lambda, c = 1 - \lambda$$

Mid point of AA' lies on $x - y - z = 1$

$$1 \Rightarrow \lambda = 2/3$$

$$\Rightarrow a = \frac{5}{3}, b = -\frac{2}{3}, c = \frac{1}{3}$$

V01/5/197. Match the following column :

Column-I

(a) If $\vec{a}, \vec{b}$ and $\vec{c}$ are three mutually perpendicular vectors where

$$|\vec{a}| = |\vec{b}| = 2, |\vec{c}| = 1, \text{ then } [\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] \text{ is}$$

Column-II

(p) -12

- (b) If $\vec{a}$ and $\vec{b}$ are two unit vectors inclined at $\frac{\pi}{3}$, then (q) 0

$$16[\vec{a} \vec{b} + \vec{a} \times \vec{b} \vec{b}] \text{ is}$$

- (c) If $\vec{b}$ and $\vec{c}$ are orthogonal unit vectors and $\vec{b} \times \vec{c} = \vec{a}$, (r) 16
then $[\vec{a} + \vec{b} + \vec{c} \vec{a} + \vec{b} \vec{b} + \vec{c}]$ is

- (d) If $[\vec{x} \vec{y} \vec{a}] = [\vec{x} \vec{y} \vec{b}] = [\vec{a} \vec{b} \vec{c}] = 0$, each vector being a non-zero (s) 1
vector, then $[\vec{x} \vec{y} \vec{c}]$ is (t) 4

Choose the correct option :

निम्न कॉलम का सुमेलित कीजिए :

स्तम्भ-I

स्तम्भ-II

- (a) यदि $\vec{a}, \vec{b}$ तथा $\vec{c}$ परस्पर लम्बवत् सदिश हैं जहाँ (p) -12

$$|\vec{a}| = |\vec{b}| = 2, |\vec{c}| = 1, \text{ तब } [\vec{a} \times \vec{b} \vec{b} \times \vec{c} \vec{c} \times \vec{a}] \text{ होगा}$$

- (b) यदि $\vec{a}$ तथा $\vec{b}$ दो इकाई सदिश हैं जो एक-दूसरे से $\frac{\pi}{3}$ कोण पर झुके हैं, (q) 0

$$\text{तो } 16[\vec{a} \vec{b} + \vec{a} \times \vec{b} \vec{b}] \text{ का मान होगा}$$

- (c) यदि $\vec{b}$ तथा $\vec{c}$ लम्बवत् इकाई सदिश हैं तथा $\vec{b} \times \vec{c} = \vec{a}$, (r) 16
तो $[\vec{a} + \vec{b} + \vec{c} \vec{a} + \vec{b} \vec{b} + \vec{c}]$ का मान होगा

- (d) यदि $[\vec{x} \vec{y} \vec{a}] = [\vec{x} \vec{y} \vec{b}] = [\vec{a} \vec{b} \vec{c}] = 0$, जहाँ सभी सदिश अशून्य हैं (s) 1
तो $[\vec{x} \vec{y} \vec{c}]$ होगा (t) 4

सही विकल्प का चयन कीजिए :

- (A) $a \rightarrow p; b \rightarrow q; c \rightarrow r; d \rightarrow s;$ (B) $a \rightarrow r; b \rightarrow p; c \rightarrow s; d \rightarrow q;$
(C) $a \rightarrow s; b \rightarrow r; c \rightarrow t; d \rightarrow p;$ (D) $a \rightarrow t; b \rightarrow s; c \rightarrow q; d \rightarrow r;$

Ans. B

Sol. (A) If $\vec{a}, \vec{b}$ and $\vec{c}$ are mutually perpendicular, then $[\vec{a} \times \vec{b} \vec{b} \times \vec{c} \vec{c} \times \vec{a}] = [\vec{a} \vec{b} \vec{c}]^2 = (|\vec{a}| |\vec{b}| |\vec{c}|)^2 = 16$

(B) Given $\vec{a}$ and $\vec{b}$ are two unit vectors, i.e., $|\vec{a}| = |\vec{b}| = 1$ and angle between them is $\frac{\pi}{3}$.

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|} \Rightarrow \sin \frac{\pi}{3} = |\vec{a} \times \vec{b}| ; \frac{\sqrt{3}}{2} = |\vec{a} \times \vec{b}|$$

$$\text{Now } [\vec{a} \vec{b} + \vec{a} \times \vec{b} \vec{b}] = [\vec{a} \vec{b} \vec{b}] + [\vec{a} \vec{a} \times \vec{b} \vec{b}] = 0 + [\vec{a} \vec{a} \times \vec{b} \vec{b}] = (\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{a}) = -|\vec{a} \times \vec{b}|^2 = -\frac{3}{4}$$

(C) If $\vec{b}$ and $\vec{c}$ are orthogonal, $\vec{b} \cdot \vec{c} = 0$.

Also, it is given that $\vec{b} \times \vec{c} = \vec{a}$.

$$\begin{aligned} \text{Now } [\vec{a} + \vec{b} + \vec{c} \vec{a} + \vec{b} \vec{b} + \vec{c}] &= [\vec{a} \vec{a} + \vec{b} \vec{b} + \vec{c}] + [\vec{b} + \vec{c} \vec{a} + \vec{b} \vec{b} + \vec{c}] \\ &= [\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{a} \cdot \vec{a} = |\vec{a}|^2 = 1 \text{ (because } \vec{a} \text{ is a unit vector)} \end{aligned}$$

(D) $[\vec{x} \vec{y} \vec{a}] = 0$

Therefore, $\vec{x}, \vec{y}$ and $\vec{a}$ are coplanar.

$$[\vec{x}, \vec{y} \vec{b}] = 0$$

Therefore, $\vec{x}, \vec{y}$ and $\vec{b}$ are coplanar.

$$\text{Also, } [\vec{a} \vec{b} \vec{c}] = 0$$

Therefore, $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar.

From (i), (ii) and (iii).

$\vec{x}, \vec{y}$ and $\vec{c}$ are coplanar. Therefore, $[\vec{x}, \vec{y} \vec{c}] = 0$

V01/5/198. Match the following column :

Column-I

Column-II

(a) Let volume of a tetrahedron ABCD is $\frac{81}{2}$ cube unit and volume (p) 9

of parallelopiped whose three coterminous edges are line segments joining centroid of any face of tetrahedron with centroids of its other three faces is V cubic unit, then V is

(b) If image of the point (1, 0, 1) in the plane (q) 8

$x - y - z = 1$ is (a,b,c) then $3(a - b + c)$ is equal to

(c) Locus of all the points which are at a distance of 3 units from the (r) 3

line $\vec{r} = \lambda(\vec{i} + \vec{j} + \vec{k})$ is given by $x^2 + y^2 + z^2 - xy - yz = \frac{k}{2}$,

then find $\frac{k}{9}$

(d) $\vec{a}, \vec{b}$ and $\vec{c}$ are such that $|\vec{a}| = \sqrt{3}, |\vec{b}| = 2$ and $|\vec{c}| = \sqrt{6}$. If $\vec{a} \cdot \vec{b} < 0; \vec{c}$ (s) 7

is perpendicular to both $\vec{a}$ and $\vec{b}$ and $\vec{a}, \vec{b}, \vec{c}$ form the coterminous edges of a tetrahedron of unit volume, then angle between $\vec{a}$ and $\vec{b}$

is $\frac{p\pi}{q}$, (where p, q are coprime numbers), where (p + q) is

Choose the correct option :

निम्न स्तम्भों का सुमेलित कीजिए :

स्तम्भ-I

स्तम्भ-II

(a) माना एक चतुष्फलक ABCD का आयतन $\frac{81}{2}$ घन इकाई है तथा

(p) 9

चतुष्फलक जिसकी सह-आरम्भिक भुजाएँ दिए गए चतुष्फलक के किसी एक फलक के केन्द्रक को अन्य तीन फलकों के केन्द्रकों से मिलाने पर बनती है, का आयतन V घन इकाई है, तो V का मान होगा

(b) यदि (1, 0, 1) का समतल $x - y - z = 1$ में प्रतिबिम्ब

(q) 8

(a, b, c) है तो $3(a - b + c)$ का मान होगा

(c) रेखा $\vec{r} = \lambda(\vec{i} + \vec{j} + \vec{k})$ से सदैव 3 इकाई दूरी पर रहने वाले

(r) 3

सभी बिन्दुओं का बिन्दुपथ $x^2 + y^2 + z^2 - xy - yz = \frac{k}{2}$ द्वारा दिया जाता है,

तो $\frac{k}{9}$ का मान ज्ञात कीजिए

(d) $\vec{a}, \vec{b}$ तथा $\vec{c}$ इस प्रकार हैं कि $|\vec{a}| = \sqrt{3}, |\vec{b}| = 2$ तथा $|\vec{c}| = \sqrt{6}$ । यदि

(s) 7

$\vec{a} \cdot \vec{b} < 0$; तथा $\vec{c}$ दोनों सदिशों $\vec{a}$ तथा $\vec{b}$ पर लम्बवत् है एवं $\vec{a}, \vec{b}, \vec{c}$

इकाई आयतन वाले चतुष्फलक की सह-आरम्भिक भुजाएँ हैं तो

$\vec{a}$ तथा $\vec{b}$ के मध्य कोण $\frac{p\pi}{q}$ है, (जहाँ p, q सह-अभाज्य संख्याएँ हैं),

तो (p + q) होगा

सही विकल्प का चयन कीजिए :

(A) $a \rightarrow p; b \rightarrow q; c \rightarrow r; d \rightarrow s$

(B) $a \rightarrow s; b \rightarrow r; c \rightarrow q; d \rightarrow p$

(C) $a \rightarrow p; b \rightarrow s; c \rightarrow r; d \rightarrow q$

(D) $a \rightarrow p; b \rightarrow q; c \rightarrow s; d \rightarrow r$

Ans. A

Sol. (P) Let $D(\vec{O}), A(\vec{a}), B(\vec{b}), C(\vec{c})$ given volume of tetrahedron is $\left| \frac{1}{6} \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix} \right| = \frac{81}{2} |(\vec{a} \vec{b} \vec{c})| = 3^5$

Centroids by faces are $G_1\left(\frac{\vec{a}+\vec{b}+\vec{c}}{3}\right), G_2\left(\frac{\vec{a}+\vec{b}}{3}\right), G_3\left(\frac{\vec{b}+\vec{c}}{3}\right), G_4\left(\frac{\vec{c}+\vec{a}}{3}\right)$

Volume of parallelopiped = $\left| \left[\overrightarrow{G_1G_2} \overrightarrow{G_1G_3} \overrightarrow{G_1G_4} \right] \right|$

$$= \left| \left[-\frac{\vec{c}}{3} - \frac{\vec{a}}{3} - \frac{\vec{b}}{3} \right] \right| = \left| \frac{[\vec{a} \vec{b} \vec{c}]}{3^3} \right| = 9$$

(Q) Let the image of the point (1, 0, 1) in the plane $x - y - z$ is A' (a, b, c)

D.R. of AA' are (a - 1, b, c - 1)

D.R. of normal are (1, -1, -1)

$$\frac{a-1}{1} = \frac{b}{-1} = \frac{c-1}{-1} = \lambda \Rightarrow a = \lambda + 1, b = -\lambda, c = 1 - \lambda$$

mid point of AA' lie on plane $x - y - z = 1 \Rightarrow \lambda = \frac{2}{3}$

$$a = \frac{5}{3}, b = -\frac{2}{3}, c = \frac{1}{3} \Rightarrow 3(a - b + c) = 3\left(\frac{8}{3}\right) = 8$$

(R) $\overrightarrow{QP} \cdot (1 + j + k) = 0 \Rightarrow (x - \lambda) + (y - \lambda) + (z - \lambda) = 0$

$$\lambda = \frac{x + y + z}{3}$$

$$(PQ)^2 = 9 \Rightarrow (x - \lambda)^2 + (y - \lambda)^2 + (z - \lambda)^2 = 9$$

$$x^2 + y^2 + z^2 + 3\lambda^2 - 2\lambda \cdot 3\lambda = 9$$

$$\Rightarrow x^2 + y^2 + z^2 - 3\left(\frac{x + y + z}{3}\right)^2 = 9$$

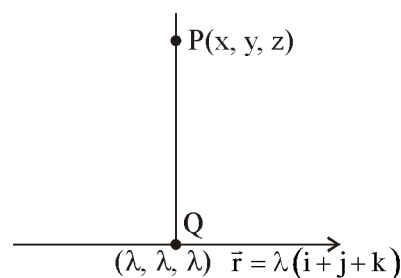
$$\Rightarrow x^2 + y^2 + z^2 - xy - yz - zx = \frac{27}{2}$$

$$k = 27, \frac{k}{9} = 3$$

(S) $\vec{C} = \lambda(\vec{a} \times \vec{b})$ also $\left| \frac{1}{6} [\vec{a} \vec{b} \vec{C}] \right| = 1 \Rightarrow |[\vec{a} \vec{b} \vec{C}]| = 6$

$$\Rightarrow |(\vec{a} \times \vec{b}) \cdot \vec{C}| = 6 \Rightarrow \frac{|\vec{C}|^2}{|\lambda|} = 6 \Rightarrow 6 = 6|\lambda| \Rightarrow |\lambda| = 1$$

$$\therefore |\vec{C}|^2 = \lambda^2 |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta$$



$$\Rightarrow 6 = 12 \sin^2 \theta = \theta = \frac{3\pi}{4} (a \cdot b < 0) p + q = 7$$

V01/5/199. Match the following column :

Column-I

Column-II

(a) If $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + \hat{k}, \vec{c} = \hat{i} + 2\hat{j} - \hat{k}$, then the value of

(p) 9

$$\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} \text{ equal to}$$

(b) If points P, Q and R have position vectors, $\vec{r}_1 = 3\hat{i} - 2\hat{j} - \hat{k}$,

(q) 28

$\vec{r}_2 = \hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{r}_3 = 2\hat{i} + \hat{j} - 2\hat{k}$ respectively, relative to an origin O, then distance of P from the plane OQR is

(c) A line with direction ratios (2, 1, 2) intersects the lines

(r) 16

$\vec{r} = -\hat{i} + \lambda(\hat{i} + \hat{j} - \hat{k})$ and $\vec{r} = -\hat{i} + \mu(2\hat{i} + \hat{j} + \hat{k})$ at A and B, then (AB) is equal to

(d) In a regular tetrahedron, the centres of the four faces are the vertices of a smaller tetrahedron. The ratio of the volume of the smaller

(s) 3

tetrahedron to that of the larger is $\frac{m}{n}$, where m and n are relatively

prime positive integers. The value of (m + n) is

Choose the correct option :

निम्न स्तम्भों को सुमेलित कीजिए :

स्तम्भ-I

स्तम्भ-II

(a) यदि $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + \hat{k}, \vec{c} = \hat{i} + 2\hat{j} - \hat{k}$, तब

(p) 9

$$\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} \text{ का मान बराबर होगा}$$

(b) यदि बिन्दुओं P, Q तथा R के स्थिति सदिश क्रमशः $\vec{r}_1 = 3\hat{i} - 2\hat{j} - \hat{k}$,

(q) 28

$\vec{r}_2 = \hat{i} + 3\hat{j} + 4\hat{k}$ तथा $\vec{r}_3 = 2\hat{i} + \hat{j} - 2\hat{k}$ हैं, जो मूल बिन्दु O के सापेक्ष हैं, तो बिन्दु P की समतल OQR से दूरी होगी

(c) दिक्अनुपात (2, 1, 2) वाली एक रेखा, रेखाओं $\vec{r} = -\hat{i} + \lambda(\hat{i} + \hat{j} - \hat{k})$ तथा

(r) 16

$\vec{r} = -\hat{i} + \mu(2\hat{i} + \hat{j} + \hat{k})$ को बिन्दु A तथा B पर प्रतिच्छेद करती है,

तब (AB) का मान होगा

- (d) एक समचतुष्फलक के चार फलकों के केन्द्रों को मिलाकर एक छोटा चतुष्फलक बनाया जाता है। यदि छोटे व बड़े चतुष्फलकों के

आयतनों का अनुपात $\frac{m}{n}$ है, जहाँ m तथा n सहअभाज्य धनात्मक पूर्णांक हैं।

(m + n) का मान होगा

सही विकल्प का चयन कीजिए :

- (A) (a) $\rightarrow$ (p), (b) $\rightarrow$ (r), (c) $\rightarrow$ (q), (d) $\rightarrow$ (s)
 (B) (a) $\rightarrow$ (q), (b) $\rightarrow$ (p), (c) $\rightarrow$ (r), (d) $\rightarrow$ (s)
 (C) (a) $\rightarrow$ (s), (b) $\rightarrow$ (q), (c) $\rightarrow$ (p), (d) $\rightarrow$ (r)
 (D) (a) $\rightarrow$ (r), (b) $\rightarrow$ (s), (c) $\rightarrow$ (s), (d) $\rightarrow$ (q)

Ans. D

Sol.

$$(A). [\vec{a} \quad \vec{b} \quad \vec{c}]^2 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix} = 16$$

$$(B). \vec{n} = \vec{r}_2 \times \vec{r}_3 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 4 \\ 2 & 1 & -2 \end{vmatrix} = \hat{i}(-6-4) - \hat{j}(-2-8) + \hat{k}(1-6) = -10\hat{i} - 5\hat{k} \Rightarrow \vec{n} = \frac{2\hat{i} - 2\hat{j} + \hat{k}}{3}$$

$$\therefore d = \vec{n} = \left| \frac{(3\hat{i} - 2\hat{j} + \hat{k}) \cdot (2\hat{i} - 2\hat{j} + \hat{k})}{3} \right| = \frac{6+4-1}{3} = 3 \text{ units}$$

$$(C). L_1 : \frac{x-1}{1} = \frac{y+1}{1} = \frac{z-0}{1} = \lambda : L_2 : \frac{x+1}{2} = \frac{y-0}{1} = \frac{z-0}{1} = \mu$$

Hence any point on L_1 and L_2 can be $(\lambda, \lambda - 1, \lambda)$ and $(2\mu - 1, \mu, \mu)$

$$\therefore \frac{2\mu - 1 - \lambda}{2} = \frac{\mu - \lambda + 1}{1} = \frac{\mu - \lambda}{1} \text{ solving } \mu = 1 \text{ and } \lambda = 3$$

A = (3, 2, 3) and B = (1, 1, 1)

$$(D). V_1 = \frac{1}{6}[\vec{a} \vec{b} \vec{c}] : V_s = \frac{1}{6}, \quad \frac{1}{27}[\vec{a} \quad \vec{b} \quad \vec{\alpha}]$$

$$\text{Hence } \frac{V_s}{V_1} = \frac{1}{27} = \frac{m}{n} \text{ or } \frac{n}{27} = \frac{m}{1} = k$$

$\therefore m$ and n are relatively prime $\Rightarrow k = 1, (m + n) = 28$